
Need-Sufficiency Architecture and Gap Diagnostics for the B-to-C Gap

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Abstract

The companion paper [Lasser, 2026d] proves the Aggregate B-to-C Lower Bound theorem: voluntary trade events in a capability collective carry revealed-preference inequalities that compose into a lower bound on realized capability volume vol_R . The theorem’s Part A requires that events be *above-sufficiency* and *assumption-admissible* (S1 holds, the trade is a free choice with non-degrading alternatives). This paper supplies the structural architecture that meets those preconditions and the diagnostic layers that interpret the bound’s output.

We introduce the *need-sufficiency architecture*: a poset-construction precondition that distinguishes involuntary need-satisfaction (where sacrifice magnitude reveals cost-of-access) from voluntary want-pursuit (where sacrifice reveals preference strength). Below-sufficiency sacrifices are structurally involuntary and filtered by the free-choice assumption (S1) before the observation channel operates. The sufficiency threshold establishes a *polarity boundary* for the sacrifice signal; the polarity-correct benchmark inverts cost dimensions so that vol_P -maximization aligns with the agent’s interest on both sides of the threshold; a multiplicative downstream-gate on vol_P -aggregation makes need-satisfaction a structural precondition for registering downstream capability weight. A parallel *need-access cost* diagnostic tracks the benchmarked capability surrendered for need-satisfaction (companion metric to β^{lower} , separate from the lower bound).

We then supply the diagnostic layers that interpret the measurement theorem’s output. The *gap decomposition* partitions the B-to-C gap $1 - \beta^{\text{lower}}$ into four disjoint cells—restricted, covered, dormant, residual—each with a well-defined δ term, under a poset-boundary closure hypothesis. The decomposition is computable in $O(|P| + E + M)$ time from observable inputs. A *wireheading-consistent concentration signal* derived from trade-flow Herfindahl–Hirschman Index (Schur-convex in the majorization order) provides a governance-thresholded alarm, with a Part-A category-flow variant third-party observable from public committed events. Finally, we re-characterize the *residual class*: the set of capabilities structurally outside the sacrifice channel’s reach. This revealed-observation carve differs from—and is more useful than—the classical benchmarkability-based residual class, yielding the irreducible component of the gap decomposition.

A four-phase worked example—a two-substrate collective transitioning through dormancy, commitment adoption, wireheading attack, and attack detection—demonstrates the architecture end-to-end.

1 Introduction

The companion paper [Lasser, 2026d] proves the *Aggregate B-to-C Lower Bound* theorem for the GFM framework’s exercised capability volume vol_R :

$$\text{vol}_R^{U,[W]} \geq \sum_n \Delta \text{vol}_P(X_n),$$

where the sum runs over admissible trade events in a window W whose bundles are pairwise poset-independent. The theorem converts the otherwise-unbounded B-to-C gap (the divergence between vol_P capability possession and vol_R capability exercise) into a bounded one on the subset of capability-space covered by observed sacrifice.

The theorem’s admissibility conditions are load-bearing. Part A requires S0 (valuation bounded by exercise), S1 (free choice with *non-degrading alternatives*), S2 (benchmark grounding), S4(a) (valuation additivity), and the genuine-exchange conditions of the revealed-sacrifice event definition. Part B additionally requires S3, S5, and within-bundle poset-independence. Of these, S1 is the most empirically delicate: not every observed sacrifice is voluntary. An agent paying above market rate for medicine because they are in severe pain, paying above market rate for water because they are dehydrated, or paying above market rate for shelter because they are exposed—none of these is a voluntary trade in the S1 sense. Such sacrifices have *reversed polarity*: more sacrifice means the agent is in a *worse* situation, not revealing stronger preference.

This paper contributes the structural architecture that makes S1 hold by construction on the events that enter the aggregate. It also develops the diagnostic layers that interpret what the lower bound is signalling when a low β^{lower} fires an alarm.

1.1 What this paper does

1. **Need-sufficiency architecture (Section 2).** A poset-construction precondition distinguishing needs from wants via multi-dimensional sufficiency thresholds. The sufficiency threshold is the structural *polarity boundary*: below it, sacrifice reveals cost-of-access and must be excluded from the want-pursuit aggregate; above it, sacrifice reveals preference strength and enters the companion paper’s aggregate [Lasser, 2026d]. A polarity-correct benchmark inverts cost dimensions; a multiplicative downstream-gate on vol_P -aggregation makes need-satisfaction a structural precondition for registering downstream weight; a parallel *need-access cost* diagnostic tracks the benchmarked capability surrendered for need-satisfaction.
2. **Gap decomposition (Section 3).** A four-cell partition of the B-to-C gap—restricted, covered, dormant, residual—with an exact identity $1 - \beta^{\text{lower}} = \sum \delta$ under poset-boundary closure. Supplies an $O(|P| + E + M)$ computability proposition from observable inputs. Interprets alarm firings: low β^{lower} with high δ_{covered} deficit is proxy failure on observed capabilities; low β^{lower} with high $\delta_{\text{restricted}}$ or δ_{residual} reflects structural invisibility rather than proxy failure.
3. **Wireheading-consistent concentration signal (Section 4).** The Herfindahl–Hirschman Index on trade-flow leverage provides a tractable, governance-thresholded concentration alarm that is Schur-convex in the majorization order. A Part-A category-flow variant is third-party observable from public committed events.
4. **Residual class characterization (Section 5).** A revealed-observation curve of the residual class: the set of capabilities structurally outside the sacrifice channel’s reach. This differs from the classical benchmarkability-based residual class in an operationally cleaner direction; the two classes overlap but neither contains the other, with explicit witnesses in each asymmetric difference.
5. **Worked example (Section 6).** A two-substrate (biology + silicon) collective traced through four phases—dormancy, commitment adoption, wireheading attack, attack detection—demonstrates the full architecture end-to-end.

1.2 Relationship to the companion paper

The companion paper [Lasser, 2026d] establishes the measurement theorem; this paper supplies the filter that keeps the channel clean and the diagnostic architecture that interprets its output. The division of labor:

Companion paper ([Lasser, 2026d])	This paper
Sacrifice observation model, S0–S5	Need-sufficiency architecture (S1 filter)
Aggregate B-to-C lower bound	Gap decomposition
Commitment-layer composition	Wireheading-consistent alarm
Two sacrifice channels	Residual class characterization
Non-self-balancing finding	Worked example

We refer to the companion paper’s definitions (revealed-sacrifice event, assumptions S0–S5, theorem `thm:aggregate_bound`, committed event, privacy theorem) without re-deriving them; the companion paper’s notation conventions carry over.

1.3 Dependencies on prior papers

This paper inherits the GFM framework [Lasser, 2026b,f,c,g,a,e,h] and the measurement-theoretic infrastructure of the companion paper [Lasser, 2026d]. A summary dependency table is collected in Appendix B.

2 Need-Sufficiency Architecture

The revealed-sacrifice observation channel ([Lasser, 2026d]) assumes all sacrifices have the same polarity: more sacrifice equals more revealed value. This is wrong for a class of sacrifices that are structurally involuntary. An agent who pays \$500 for insulin is not revealing that insulin is worth \$500 beyond baseline, they are revealing that baseline functionality costs \$500 to maintain. The sacrifice magnitude reveals *cost-of-access* (how expensive it is for this agent to reach minimum viable functionality), not *preference strength* (how much the agent values the outcome beyond baseline). Larger sacrifice means a worse situation, not higher revealed value.

The framework needs a structural requirement on poset construction that distinguishes need-satisfaction from want-pursuit and identifies the boundary where sacrifice polarity flips.

2.1 Three motivating observations

Needs are neither purely maximizing nor minimizing. Every need has *both* maximizing dimensions (quality, quantity, reliability) and minimizing dimensions (cost, distance, time). Needs are not a single-polarity class. What *is* true: the minimizing dimensions of needs are where the polarity problem lives, because cost-of-access is involuntary.

Scalar sufficiency hides structural defects. A single “water access: sufficient” measurement might hide that the water is contaminated. The agent has water (passes the scalar threshold) but the way the need is met actively degrades future vol_P through downstream effects on health, cognition, and reproduction. The framework reports green while the system corrodes from the root.

Downstream-safe bundles. Three levels of sufficiency, each strictly stronger:

- *Level 1: Scalar threshold (insufficient).* “Water: yes/no.” Hides quality, reliability, and cost defects.
- *Level 2: Bundle threshold (necessary but not sufficient).* All dimensions of the need must independently meet their thresholds simultaneously. This catches contamination (quality dimension fails), but misses subtler failures where the satisfaction pathway degrades downstream capabilities over time. For example, when the contamination thresholds are incorrectly tuned.
- *Level 3: Downstream-safe bundle.* Bundle thresholds are all met *and* the need-satisfaction pathway does not produce net-negative effects on capabilities in the need’s downstream capability cone.

2.2 Formal definitions

Definition 1 (Multi-Dimensional Need Bundle). *A capability N identified as a need at position p_N in the poset has:*

- **Dimensions** $\{d_1, \dots, d_m\}$ with sufficiency thresholds $\{s_1, \dots, s_m\}$.
- Each dimension has a **polarity**: maximizing (quality, quantity, reliability) or minimizing (cost, distance, time).
- **Downstream cone** $\text{Down}(p_N) = \{c : c \text{ depends on } N \text{ in the poset}\}$.
- **Satisfaction pathway** π_N : the specific way the agent meets the need (which source, at what quality, at what cost).

For cost dimensions (minimizing polarity), the benchmark inverts the natural direction: $\text{actual}_k = 1/\text{cost}_k$, with sufficiency at $s_k = 1/\text{max_acceptable_cost}$, so that benchmark maximization aligns with cost minimization.

Definition 2 (Downstream-Safe Sufficiency). Let $\Pi_N^0 = \{\pi : d_k(\pi) = s_k \text{ for all } k\}$ be the class of pathways satisfying each dimension of N at exactly its sufficiency threshold (assumed non-empty). Let $\Pi_N^{\text{ref}} \subseteq \Pi_N^0$ be a non-empty admissible reference set, designated externally by governance (Remark 1): the threshold-level pathways against which downstream non-degradation is evaluated. Need N at position p_N is downstream-safe sufficient under satisfaction pathway π_N iff:

- Bundle thresholds met**: $d_k(\pi_N) \geq s_k$ for all dimensions k (with cost dimensions inverted per Definition 1).
- Downstream non-degradation against all admissible references**: π_N dominates every $\pi^* \in \Pi_N^{\text{ref}}$ on every downstream capability simultaneously:

$$\text{quality}_c(\text{with } N \text{ satisfied via } \pi_N) \geq \text{quality}_c(\text{with } N \text{ satisfied via } \pi^*) \quad \text{for all } c \in \text{Down}(p_N) \text{ and all } \pi^* \in \Pi_N^{\text{ref}}$$

The universal quantifier over Π_N^{ref} is load-bearing: a merely existential condition (“dominate some threshold-level pathway”) would admit materially degrading pathways that happen to beat a pathologically weak reference. Under the common singleton case $\Pi_N^{\text{ref}} = \{\pi_N^{\text{ref}}\}$, the condition reduces to dominance against one canonical baseline. Each element of Π_N^{ref} must be a realizable pathway, not a coordinatewise supremum over Π_N^0 that no single pathway achieves.

Remark 1 (Governance designates the admissible reference set). The admissible reference set Π_N^{ref} is a public governance input, not a framework-computed object. Institutions typically adopt a specific reference pathway as the canonical baseline (“standard of care” in medicine, “best available technology” in environmental policy); in those settings Π_N^{ref} is a singleton. Multi-reference settings arise when different disciplines canonicalize differently and π_N must satisfy all of them simultaneously; the downstream-safety condition then requires robust dominance over every admissible baseline. Governance may update Π_N^{ref} over time as standards evolve; the framework reads the current Π_N^{ref} at evaluation time and does not attempt to re-derive it from first principles.

Remark 2 (Structural connection to exercise-pathway machinery). The need-satisfaction pathway π_N is structurally an exercise pathway in the sense of [Lasser, 2026c, Definition 2]: a sequence of capability invocations producing an outcome (here, the satisfaction of need N). This identification lets the framework treat π_N as a first-class object rather than an abstract “way of satisfying the need.”

The downstream-safety condition (Definition 2(b)) is, however, a distinct mathematical object from the concentration-risk measure of [Lasser, 2026c, Definition 3]. Concentration risk is a scalar worst-case contraction estimate over an entire pathway; downstream-safety is a per-capability quality comparison $\text{quality}_c(\pi_N) \geq \text{quality}_c(\pi^*)$ against every reference pathway $\pi^* \in \Pi_N^{\text{ref}}$, indexed by $c \in \text{Down}(p_N)$. Neither reduces to the other. Evaluating $\text{quality}_c(\pi_N)$ in practice requires either (i) direct observation of c ’s performance conditional on the pathway, or (ii) structural inference from the poset’s cooperative structure. A full formalisation of the evaluation procedure is outside this paper’s scope; within the paper, Definition 2(b) is a structural condition the framework enforces (a pathway failing the condition is rejected as a satisfaction pathway for N), not a quantity the framework computes directly.

A policy consequence, not a theorem: because need-satisfaction pathways are not optional (the agent must satisfy the need), the framework’s response to a pathway failing Definition 2(b) differs from its response to a risk claim under [Lasser, 2026h]. A failing risk pathway is restricted (the capability is blocked). A failing satisfaction pathway must be replaced (the need must still be met, but

through a pathway satisfying Definition 2). This governance distinction is upstream of the paper’s formal machinery; it names the correct operational response when the condition fails, not a new computation the framework performs.

Definition 3 (Sufficiency Threshold as Polarity Boundary). *The sufficiency bundle is the structural boundary where revealed-sacrifice polarity flips:*

- **Below sufficiency on any dimension:** *The agent is investing to reach minimum viable functionality. Sacrifice is involuntary (the agent must reach sufficiency to operate). Sacrifice magnitude reveals **cost-of-access**—how expensive it is for this agent to reach baseline on this dimension. Larger sacrifice = worse situation.*
- **Above sufficiency on all dimensions:** *The agent has reached functionality and is choosing to invest further. Sacrifice is voluntary (the agent could stop at “good enough”). Sacrifice magnitude reveals **preference strength**—how much the agent values improvement beyond baseline. Larger sacrifice = higher revealed value.*

The sufficiency threshold is where the need-insufficiency failure mode of Assumption S1 ([Lasser, 2026d]) is removed: below sufficiency, S1 fails because retaining the sacrificed capability is not a non-degrading alternative; above sufficiency, this specific failure mode does not apply (though S1 can still fail via other mechanisms—see Proposition 3, Remark).

Definition 4 (Polarity-Correct Benchmark). *For each dimension d_k of a need capability:*

- **Below sufficiency:** $\text{benchmark}_k = \min(\text{actual}_k/s_k, 1)$. Climbs from 0 toward 1 as the agent approaches sufficiency. Caps at 1. Maximizing this equals closing the sufficiency gap.
- **Above sufficiency:** $\text{benchmark}_k = 1 + \alpha_k \cdot \max((\text{actual}_k - s_k)/s_k, 0)$, where $\alpha_k < 1$ discounts above-sufficiency gains relative to below-sufficiency gains. This makes closing the sufficiency gap more valuable per unit of improvement than enhancing beyond sufficiency—the correct incentive structure for needs.

Definition 4 operates at the single-capability level: it determines the benchmark score for each dimension of a given capability. It does not, by itself, model the structural fact that need-satisfaction is a precondition for the exercise of downstream capabilities: an agent without water does not merely lose one bit of water-access capability, they lose access to the entire class of cooperative capabilities that require a hydrated body. To capture this, the framework imposes a multiplicative gate on vol_P-aggregation through the downstream cone.

Definition 5 (Downstream-Gated Capability Weighting). *For each capability $c \in P$, let $\mathcal{N}(c) = \{N : c \in \text{Down}(p_N)\}$ be the set of needs whose downstream cones contain c . For each such N , let $s_N(c) \in [0, 1]$ be the sufficiency level of N in the context where c is exercised, with $s_N(c) = 1$ when N is downstream-safe sufficient (Definition 2) and $s_N(c) = 0$ when N is fully unmet. In the common case where sufficiency is uniform across the downstream cone, we abbreviate $s_N(c) \equiv s_N$ and treat s_N as a scalar.*

The effective weight of c is:

$$w^{\text{eff}}(c) = w(c) \cdot \prod_{N \in \mathcal{N}(c)} G(s_N(c)),$$

where $G : [0, 1] \rightarrow [0, 1]$ is a gate function satisfying: $G(1) = 1$, $G(0) = \alpha_0 \in [0, 1]$ (the gate’s floor), G monotone non-decreasing, and G right-continuous at $s = 0$ (so $\lim_{s \rightarrow 0^+} G(s) = \alpha_0$). A gate with $\alpha_0 = 0$ is called a zero-floor gate; a gate with $\alpha_0 > 0$ is called a positive-floor gate. The four candidate gates of Remark 3 all satisfy right-continuity; step-function gates are excluded. Linear, ReLU, and power-law gates are zero-floor; leaky-ReLU with parameter $\alpha > 0$ is positive-floor with $\alpha_0 = \alpha$.

The effective capability volume is:

$$\text{vol}_P^{\text{eff}}(G) = \sum_{c \in P} w^{\text{eff}}(c).$$

The framework optimizes $\text{vol}_P^{\text{eff}}$ rather than raw vol_P once needs have been identified.

Remark 3 (Candidate gate functions). *The choice of G is a design parameter that trades off smoothness, computational simplicity, and the strength of need-dominance. Four candidates:*

- **Linear:** $G(s) = s$. Simplest; gentle need-dominance; $s_N(c) = 0.5$ halves the downstream contribution.
- **Leaky-ReLU:** $G(s) = \alpha$ for $s \leq s_0$, then linear ramp from α at s_0 to 1 at $s = 1$. Parameters: threshold $s_0 \in [0, 1)$, leak $\alpha \in [0, 1)$. Below threshold, downstream capabilities retain a constant floor α ; above, rapid ramp to full contribution.
- **ReLU:** Leaky-ReLU with $\alpha = 0$. Below threshold, downstream capabilities are fully suppressed; above, linear ramp. Hardest gate; strongest need-dominance.
- **Power law:** $G(s) = s^\gamma$ for $\gamma \geq 1$. Smooth everywhere; gradient vanishes at $s = 0$ for $\gamma > 1$, giving progressively stronger penalty as the sufficiency gap widens. Single tunable parameter.

The appropriate gate shape depends on the domain. Biological needs with cliff-like failure modes (dehydration, oxygen deprivation) warrant ReLU or high- γ power law. Needs with graceful degradation (social connection, mild nutritional deficit) warrant leaky-ReLU or low- γ power law. The paper leaves G as a domain-specific design parameter and states subsequent results for any valid G .

Proposition 1 (Near-Total Collapse Under Unmet Need). *Let G be any valid gate function (Definition 5) with floor $\alpha_0 = G(0) \in [0, 1)$, and suppose $s_N(c) \equiv s_N$ for all $c \in \text{Down}(p_N)$ (homogeneous sufficiency). Partition the poset as $P = \text{Down}(p_N) \sqcup \overline{\text{Down}(p_N)}$, with weight sums $W_{\text{down}} = \sum_{c \in \text{Down}(p_N)} w(c)$ and $W_{\text{out}} = \sum_{c \in \overline{\text{Down}(p_N)}} w(c)$, so $\text{vol}_P^{\text{raw}} = W_{\text{down}} + W_{\text{out}}$. If the agent is at sufficiency level $s_N < 1$:*

$$\text{vol}_P^{\text{eff}}(G) \leq G(s_N) \cdot W_{\text{down}} + W_{\text{out}}.$$

Collapse limits. As $s_N \rightarrow 0$, $G(s_N) \rightarrow \alpha_0$ by right-continuity:

- Zero-floor gate ($\alpha_0 = 0$): the downstream contribution vanishes, $\text{vol}_P^{\text{eff}} \rightarrow W_{\text{out}}$. When $W_{\text{down}} \gg W_{\text{out}}$, the effective volume collapses to a small fraction of its unrestricted value.
- Positive-floor gate ($\alpha_0 > 0$): the downstream contribution collapses to its floor, $\text{vol}_P^{\text{eff}} \rightarrow \alpha_0 W_{\text{down}} + W_{\text{out}}$. The effective volume retains a positive fraction α_0 of downstream weight even under fully unmet need.

The multiplicative structure makes need-satisfaction a precondition for vol_P to register the full downstream portion of the agent's capabilities; an agent cannot maximize $\text{vol}_P^{\text{eff}}$ without closing sufficiency gaps on needs whose downstream cones cover their capability-weight. Whether unmet need collapses the downstream contribution entirely or only to the floor $\alpha_0 W_{\text{down}}$ is a design choice governed by the gate's floor parameter.

Proof. Split the sum over P by membership in $\text{Down}(p_N)$. For $c \in \text{Down}(p_N)$, the effective weight is $w^{\text{eff}}(c) = w(c) \cdot G(s_N) \cdot \prod_{N' \in \mathcal{N}(c), N' \neq N} G(s_{N'}(c)) \leq w(c) \cdot G(s_N)$ since each additional gate factor is ≤ 1 . For $c \in \overline{\text{Down}(p_N)}$, $N \notin \mathcal{N}(c)$, so $w^{\text{eff}}(c) = w(c) \cdot \prod_{N' \in \mathcal{N}(c)} G(s_{N'}(c)) \leq w(c)$. Summing the two cases:

$$\text{vol}_P^{\text{eff}} \leq G(s_N) \cdot \sum_{c \in \text{Down}(p_N)} w(c) + \sum_{c \in \overline{\text{Down}(p_N)}} w(c) = G(s_N) \cdot W_{\text{down}} + W_{\text{out}}.$$

Right-continuity of G at 0 gives $\lim_{s_N \rightarrow 0^+} G(s_N) = G(0) = \alpha_0$, so the downstream term tends to $\alpha_0 W_{\text{down}}$: zero for zero-floor gates, the floor value for positive-floor gates. ■

Remark 4 (The non-downstream floor and the gate floor). *Two floors appear at $s_N = 0$: the always-present non-downstream floor W_{out} (capabilities not depending on N , which no gate can suppress), and the gate-dependent downstream floor $\alpha_0 W_{\text{down}}$ (which is zero for zero-floor gates and $\alpha \cdot W_{\text{down}}$ for positive-floor gates).*

for leaky-ReLU with leak $\alpha > 0$). The worked table in Remark 6 exhibits both: at $s_N = 0$, linear, ReLU, and power-law gates give $0.231 = W_{\text{out}}/\text{volp}^{\text{raw}} = 3/13$ (only the non-downstream floor), while leaky-ReLU with $\alpha = 0.1$ gives $0.308 = (3+10 \cdot 0.1)/13 = (W_{\text{out}} + \alpha_0 W_{\text{down}})/\text{volp}^{\text{raw}}$ (both floors). The proposition’s “total collapse” force applies to zero-floor gates with $W_{\text{down}} \gg W_{\text{out}}$; for positive-floor gates, the collapse is bounded below by the gate’s floor regardless of $W_{\text{down}}/W_{\text{out}}$ ratio.

Proposition 2 (Need-Dominance in the Gradient). *Assume homogeneous sufficiency $s_N(c) \equiv s_N$ for all $c \in \text{Down}(p_N)$ and that all other needs are at sufficiency ($s_{N'}(c) = 1$ for $N' \neq N$), so that s_N is a scalar. For any valid gate function G differentiable at s_N with $G'(s_N) > 0$, the two relevant partial derivatives of volp^{eff} are*

$$\frac{\partial \text{volp}^{\text{eff}}}{\partial s_N} = G'(s_N) \cdot W_{\text{down}}(N), \quad \frac{\partial \text{volp}^{\text{eff}}}{\partial w(c')} = \begin{cases} 1 & c' \notin \text{Down}(p_N), \\ G(s_N) \leq 1 & c' \in \text{Down}(p_N), \end{cases}$$

where $W_{\text{down}}(N) = \sum_{c \in \text{Down}(p_N)} w(c)$. Both derivatives are rates on the same unit-step scale (one unit of s_N vs. one unit of $w(c')$, treating s_N and w as commensurable intervention variables; see Remark 5).

Marginal-dominance condition. *Under the common intervention scale, the marginal return to closing the sufficiency gap on N exceeds the marginal return to unit-weight addition on any single capability whenever*

$$G'(s_N) \cdot W_{\text{down}}(N) > 1.$$

For linear gates ($G(s) = s$), this reduces to $W_{\text{down}}(N) > 1$. For leaky-ReLU or ReLU gates, the condition holds at any s_N in the ramp region provided $W_{\text{down}}(N)$ exceeds the reciprocal of the local slope. For power-law gates $G(s) = s^\gamma$ with $\gamma > 1$, the condition fails near $s_N = 0$ (where $G'(s_N)$ is small) even for large downstream cones—an intentional feature of gates that weight severity of unmet need over its breadth.

Proof. Under the stated hypothesis, every gate product collapses: for $c \in \text{Down}(p_N)$, the product $\prod_{N' \in \mathcal{N}(c)} G(s_{N'}(c))$ equals $G(s_N) \cdot \prod_{N' \neq N} G(1) = G(s_N)$. Thus $w^{\text{eff}}(c) = w(c) \cdot G(s_N)$ for $c \in \text{Down}(p_N)$ and $w^{\text{eff}}(c) = w(c)$ otherwise. Summing gives $\text{volp}^{\text{eff}} = G(s_N) \cdot W_{\text{down}}(N) + W_{\text{out}}(N)$. Only the first term depends on s_N , so differentiation yields $\partial \text{volp}^{\text{eff}} / \partial s_N = G'(s_N) \cdot W_{\text{down}}(N)$. Differentiating in $w(c')$: the capability c' ’s gated weight is $G(s_N) \cdot w(c')$ if $c' \in \text{Down}(p_N)$ and $w(c')$ otherwise, giving the stated partial derivatives. Comparing the two partials on the common unit-step scale yields the dominance condition. ■

Remark 5 (Finite-step form and intervention-cost scale). *The marginal-dominance condition is a per-unit-cost comparison under the convention that one unit of Δs_N and one unit of Δw have the same intervention cost. Its finite-step form is the Taylor expansion: a sufficiency increment $\Delta s_N = \delta$ produces gain $[G(s_N + \delta) - G(s_N)] \cdot W_{\text{down}}(N) = G'(s_N) W_{\text{down}}(N) \cdot \delta + O(\delta^2)$, while a weight increment $\Delta w(c') = \delta$ on a single capability produces gain $\delta \cdot \max(1, G(s_N)) \leq \delta$. For small enough δ , the marginal condition $G'(s_N) W_{\text{down}}(N) > 1$ implies the sufficiency increment strictly dominates; a finite δ for which $[G(s_N + \delta) - G(s_N)] \cdot W_{\text{down}}(N) > \delta$ is the exact condition. The condition $G'(s_N) W_{\text{down}}(N) > 1$ is not a claim that a finite Δs_N of arbitrary size beats a finite unit of weight: e.g., with $G(s) = s$ and $W_{\text{down}} = 10$, a $\Delta s_N = 0.01$ gain of 0.1 does not exceed a unit-weight gain of 1 in absolute terms—only in per-unit-of-intervention terms (the sufficiency increment returned ten times the per-unit gain). Interpreting the condition as anything other than a marginal-rate comparison would conflate the two intervention scales.*

Remark 6 (Worked comparison across candidate gates). *Consider an agent with 10 cooperative capabilities in $\text{Down}(p_N)$ (weight 1 each) and 3 capabilities outside the cone (weight 1 each), so $\text{volp}^{\text{raw}} = 13$. At homogeneous sufficiency level $s_N(c) \equiv s$ for all $c \in \text{Down}(p_N)$, $\text{volp}^{\text{eff}}(s) = 3 + 10 \cdot G(s)$. The following table compares effective volume as a fraction of volp^{raw} for the four candidate gates (Leaky-ReLU with $\alpha = 0.1$, $s_0 = 0.8$; ReLU with $s_0 = 0.8$; power law with $\gamma = 2$):*

s	Linear	Leaky-ReLU	ReLU	Power ($\gamma=2$)	Power ($\gamma=3$)
0.00	0.231	0.308	0.231	0.231	0.231
0.25	0.423	0.308	0.231	0.279	0.243
0.50	0.615	0.308	0.231	0.423	0.327
0.75	0.808	0.308	0.231	0.663	0.556
0.80	0.846	0.308	0.231	0.723	0.625
0.90	0.923	0.654	0.615	0.854	0.792
1.00	1.000	1.000	1.000	1.000	1.000

Linear gives the gentlest gate (at $s = 0.5$, downstream contributions are only halved, so 62% of raw vol_P is still registered). *ReLU* gives the hardest gate (below threshold $s_0 = 0.8$, downstream contributions are fully suppressed; $\text{vol}_P^{\text{eff}}$ stays at the non-downstream floor of $3/13 \approx 23\%$). Power-law gates sit between, with higher γ approaching ReLU’s sharpness. Leaky-ReLU preserves the ReLU shape above threshold while leaving a small floor α below.

2.3 Sacrifice-polarity (S1) connection

Proposition 3 (S1–Sufficiency Connection). *Below-sufficiency sacrifices fail Assumption S1 (free choice, [Lasser, 2026d]). S1 requires that retaining the sacrificed capability X_n was a non-degrading alternative—one that would not leave the agent below minimum viable functionality on any need dimension. For an agent whose capability level on dimension d_k is below the sufficiency threshold s_k , retaining X_n (not making the trade) leaves the agent below functionality on d_k . This is not a non-degrading alternative per S1’s definition, so S1 fails for that event.*

Consequence for the sacrifice channel. Below-sufficiency events are filtered from the revealed-sacrifice aggregate by S1 before they reach the aggregate lower bound (the aggregate lower bound theorem [Lasser, 2026d]). The sacrifice channel operates only over events where S1 holds. For above-sufficiency (want-pursuit) events, the need-insufficiency failure mode of S1 is removed. S1 may still fail for above-sufficiency events via other mechanisms (coercion, duress—see Remark 7 below).

Remark 7 (S1 is not automatically satisfied above sufficiency). *S1 can still fail for above-sufficiency events if the trade was coerced by other mechanisms (economic duress, political pressure, information asymmetry) even though the need-sufficiency condition is met. The sufficiency threshold resolves one structural failure mode of S1 (the involuntary-need-satisfaction case); other S1 failure modes (the coercion open question in [Lasser, 2026d], Section 8) remain.*

Proof of Proposition 3. Let (i, X, Y, t) be a sacrifice event where, absent this trade, agent i ’s capability level on need dimension d_k would fall below the sufficiency threshold s_k . Agent i must reach s_k on d_k to maintain minimum viable functionality. The alternative of retaining X (not making the trade) leaves the agent below s_k on d_k . S1 requires the retention alternative to be non-degrading: it must not drop the agent below minimum viable functionality on any need dimension. Retention here does precisely that, so the non-degrading alternative requirement fails and S1 does not hold for this event. Events with failed S1 are excluded from the aggregate by the applicable-assumptions clause of the aggregate lower bound theorem [Lasser, 2026d]. ■

2.4 The capability-stuffing problem

An agent optimizing vol_P under the non-triviality axiom M5 (every valid capability with $s_{\max} \geq 1$ contributes positive weight [Lasser, 2026f, Proposition 1]) has a natural incentive to add capabilities that pass structural validation but do not increase experiential optionality. We call this the *capability-stuffing* failure mode.

The canonical form is redundant capability construction targeting a single underlying need from multiple vantage points. “Travel from home to a gas station” is a real, benchmarkable *do*-capability: it satisfies the axioms M1–M6, has a well-defined metric ($s_{\max} \geq 1$), and represents genuine structural knowledge about the world. Adding “travel from work to a gas station,” “travel from the grocery store to a gas station,” and so on produces a sequence of further real, benchmarkable capabilities, each passing the same axiomatic checks. Collectively they inflate vol_P without broadening

what the agent can actually do: every addition instantiates the same underlying optionality (fuel access) from a different origin.

Capability-stuffing is distinct from other failure modes in the sequence. It is not *wireheading* (Lasser [2026a], Discussion), which is narrow self-reinforcing exercise of existing capabilities; stuffing adds new capabilities without exercising them degenerately. It is not the *Doll Problem* (Lasser [2026b], Section 7.1), which is possessed-but-unexercised capabilities; the stuffed capabilities may be technically exercised (routing through the gas station “exercises” the distance-measurement). It is not the *B-to-C gap* in the aggregate sense this paper otherwise treats; stuffing widens the gap through a specific mechanism (individual-weight inflation) that the aggregate lower bound of the aggregate lower bound theorem [Lasser, 2026d] does not, by itself, address. Stuffing is a framework-native perverse instantiation: the optimizer discovers it by following the vol_P -maximization gradient under M5, not by departing from the framework’s rules.

Tension with structural discovery. Capability-stuffing is closely related to a *legitimate* mechanism: the structural discovery value of [Lasser, 2026c, Proposition 2], which holds that discovering new poset structure is value-positive. An agent adding a new distance-measurement capability to its repertoire is making a structural discovery in that sense. The formal distinction between legitimate discovery and capability-stuffing turns on whether the new capability creates new cooperative structure:

- **Legitimate structural discovery** enriches the poset by creating cooperative capabilities (axiom M6 superadditivity): the new capability interacts with existing capabilities to produce outputs previously unachievable. The marginal contribution to vol_P includes cooperative terms beyond the individual weight.
- **Capability-stuffing** adds individual capabilities whose marginal cooperative contribution is zero (or effectively so): the new capability’s weight accrues through M5 alone, not through M6. No new cooperative outputs become achievable.

Without a structural mechanism that distinguishes the two, the framework rewards both equally, and a vol_P -maximizer with capability-discovery latitude will exploit the undifferentiated reward to stuff.

What the need-sufficiency architecture does resolve. The downstream-safe bundle sufficiency condition (Definition 2) and the polarity-correct benchmark (Definition 4) together handle one specific class of redundant construction: capabilities whose claimed contribution lies *below* the sufficiency threshold on an already-satisfied need dimension. For those, the below-sufficiency benchmark saturates at 1, so additional instances do not inflate the benchmark. This is a real structural property of the architecture and it is worth stating narrowly:

Proposition 4 (Below-Sufficiency Saturation (pointwise)). *For any need dimension d_k of any single capability, the below-sufficiency component of the polarity-correct benchmark (Definition 4) saturates at 1 once $\text{actual}_k \geq s_k$. This is a pointwise cap: the benchmark value assigned to dimension d_k of one capability does not exceed 1, regardless of how many times that capability’s actual_k is re-evaluated or re-instantiated.*

Proof. Below-sufficiency benchmark for dimension d_k is $\min(\text{actual}_k/s_k, 1)$, which is bounded above by 1 pointwise. ■

Remark 8 (The pointwise cap does not extend to an aggregate dimension-level cap). *Proposition 4 is a per-capability, per-dimension cap, not a dimension-level aggregate cap. Distinct capabilities $c_1, c_2, \dots$ each reporting $\text{actual}_k \geq s_k$ on the same dimension d_k each score 1 under the below-sufficiency component; their M5-weighted sum in vol_P is not capped by this proposition and accumulates with each new capability. The proposition therefore blocks only a narrow class of stuffing: re-instantiating a single capability to inflate its own benchmark via repeated reporting. The broader class of capability-stuffing—creating distinct capabilities that each satisfy sufficiency on the same dimension—is not structurally blocked and is acknowledged as unresolved below.*

What the need-sufficiency architecture does not resolve. Proposition 4 is the full extent of the architecture’s structural anti-stuffing contribution. It does *not* cover the following:

- (i) Above-sufficiency re-instantiation. An attacker can construct capabilities with distinct actual_k values exceeding s_k (e.g., home-to-gas at 15 minutes and work-to-gas at 5 minutes). Each earns positive above-sufficiency credit $\alpha_k(\text{actual}_k - s_k)/s_k$ under Definition 4, even though the underlying optionality (fuel access) is unchanged.
- (ii) Capabilities whose claimed dimension is not inside any existing need’s downstream cone, for which the gate function (Definition 5) does not apply and M5 weight accrues fully to vol_P .
- (iii) Arbitrary asymmetric decoration — distinct cooperative descendants, minor weight perturbations, unique subsumption edges — that preserves M5 credit for the stuffed capability regardless of whether the capability provides genuinely new optionality.

Each of these defeats any attempt to block stuffing using post-declaration poset properties or exercise patterns alone. The honest statement of the need-sufficiency architecture’s contribution is sacrifice-polarity resolution (via Proposition 3 above, which is load-bearing for the sacrifice channel) together with the narrow below-sufficiency cap of Proposition 4. Full structural resolution of M5 stuffing requires machinery this paper does not develop.

Capability-stuffing as structural avoidance. The failure pattern across attempts is that every signature over post-declaration data is adversarially manipulable. The move that changes the framing is recognizing stuffing as a specific instance of *structural avoidance*, the pathology flagged as an open gap in [Lasser, 2026f, Discussion]: when the framework rewards capability declarations additively, an agent optimizing vol_P has an incentive to *decline to discover* that capabilities merge, factor, or are otherwise reducible. Stuffing is the concrete manifestation: the agent keeps “travel from home to gas station” and “travel from work to gas station” distinct in P rather than merging them into the canonical form “access to fuel from multiple origins” that a proper structural discovery would produce. Each unmerged capability inflates the M5 count; the merged capability has the same optionality content.

The positive dual is already in the sequence: [Lasser, 2026c, Proposition 2] establishes that structural discovery is value-positive under conditions on the discovered structure’s information value. Making the symmetric claim—that refusal to discover is value-negative—requires mechanism, not entailment. The direction is a *proof-burden taxonomy* with *claim-and-challenge back-pressure*.

The proof-burden direction. For each avoidance route, a corresponding burden the agent must discharge to retain vol_P credit. Three types: proof-of-non-mergeability (for the stuffing-specific case), proof-of-provenance (blocking declarations without underlying structural work), and proof-of-use (blocking declarations that remain persistently dormant outside the residual class R_S of §5). Proof-burdens operate as preprocessing on the poset before M5 is applied; the M5 formula is preserved, but the measure is evaluated on the canonicalized poset rather than the presented one. Canonicalization itself is new axiomatic content at the preprocessing layer.

Claims are temporal and contestable. An agent’s proof-of-non-mergeability stands until a counter-argument demonstrates a merger mechanism; the discoverer of that mechanism receives structural-discovery credit ([Lasser, 2026c, Proposition 2]); the avoider loses the M5 credit whose proof has been voided. This makes avoidance economically risky rather than economically stable. Analogous architectures exist in scientific consensus, patent challenge, and proof-of-stake slashing; adapting any of them requires substantive protocol and calibration work.

Scope of the follow-up. The sequence provides substrate for parts of this construction: Paper 5’s commitment protocol for timestamped claims, Paper 8’s sacrifice channel for partial use-evidence, Paper 6’s controlled relaxation as an adjudication pattern. Substrate is not implementation: formalizing the canonicalization operator, the three proof-burden types with admissible evidence classes, the claim-and-challenge protocol with challenge-market robustness (admission filtering against spam, identity weighting against Sybil flooding, independence requirements against cartel suppression), and the equilibrium-dominance theorem requires a dedicated follow-up paper. This paper states the direction and leaves the formal work there. The open-question entry in §8 describes the follow-up scope in more detail.

2.5 How sufficiency thresholds are set

Four mechanisms, not mutually exclusive:

(a) Poset-structural. Capabilities with high fan-out (many downstream dependents) have their sufficiency level set by the minimum required for downstream capabilities to function. Computable from poset topology.

(b) Empirical / cross-sectional. Observe the population’s actual capability levels and identify the threshold below which agents demonstrably cease functioning on downstream capabilities.

(c) Revealed-sacrifice-derived. Sufficiency is the level at which sacrifice polarity flips. Below sufficiency, sacrifice is involuntary and increasing; above sufficiency, sacrifice becomes voluntary and choice-driven. The transition point in the sacrifice data identifies the threshold empirically. This is the most framework-native option but has a bootstrap problem: identifying the threshold requires classifying events, but classifying events requires the threshold. Resolution: start with structural/normative thresholds (mechanisms (a) and (d)), refine using sacrifice data.

(d) Normative / external. A human authority or governance process declares sufficiency levels. This is the approach of Sen [1999] and Nussbaum [2011]. The framework’s formal apparatus is agnostic to the source; it needs the thresholds as input.

Primary mechanism: (c) with (a) as structural validation. The revealed-sacrifice data shows where the transition from involuntary to voluntary sacrifice occurs; the poset topology confirms the transition corresponds to a structurally meaningful threshold.

3 Alarm Mechanism and Gap Decomposition

3.1 The alarm

Definition 6 (B-to-C Evidence Alarm). *The B-to-C evidence alarm fires when the lower-bound B-to-C ratio drops below a threshold:*

$$\text{ALARM}(T) = \beta^{\text{lower}}(G, T) < \beta_{\text{alarm}},$$

where $\beta_{\text{alarm}} \in (0, 1)$ is configurable. The alarm is an evidence alarm, not a proxy-failure alarm directly: when β^{lower} is low, the sacrifice channel shows little evidence of vol_R -content proportional to vol_P , but this does not by itself imply that true exercise is low. The interpretation of a firing alarm requires additional analysis (via the gap decomposition below) to determine whether low evidence reflects genuine low exercise, restricted capabilities, dormant wants, or residual-class capabilities beyond the channel’s reach.

Remark 9 (Evidence alarm versus proxy-failure diagnosis). *Because $\beta^{\text{lower}} \leq \beta_{\text{true}}$, the alarm distinguishes two regimes:*

- **Alarm fires and δ_{covered} is small relative to $\delta_{\text{dormant}} + \delta_{\text{restricted}} + \delta_{\text{residual}}$:** the low β^{lower} reflects insufficient observed trade on non-residual, non-restricted capabilities. This is the canonical proxy-failure signature: vol_P -weighted capabilities exist that are observable in principle but not being traded.
- **Alarm fires and δ_{residual} or $\delta_{\text{restricted}}$ dominates:** the low β^{lower} reflects structural invisibility or policy restriction, not proxy failure. The framework should not escalate in these cases; the alarm is firing on what the sacrifice channel cannot reach, not on degraded exercise.

The alarm has no false positives in the narrow sense—low β^{lower} is real and the sacrifice channel genuinely sees little evidence—but it has false positives in the proxy-failure sense, which is why the gap decomposition below is necessary to distinguish the regimes. False negatives (vol_R high but invisible to the sacrifice channel) are also possible and are the structural cost of privacy-minimal observation.

3.2 Diagnostic decomposition

Definition 7 (Gap Decomposition Under Revealed Sacrifice). *Under the aggregate lower bound theorem [Lasser, 2026d] Part B’s hypotheses (S0–S5), the per-capability aggregate $\text{vol}_R^{\text{lower}}(c)$ is well-defined, and subset aggregates $\text{vol}_R^{\text{lower}}(A)$ are given by the subset-extension of $\text{vol}_R^{\text{lower}}$ [Lasser, 2026d]. Under those hypotheses, the B-to-C gap $(1 - \beta^{\text{lower}})$ decomposes into four disjoint sources. Each capability in P is assigned to exactly one category by priority ordering (restricted > covered > dormant > residual), and each category’s δ term is its uncovered vol_P share:*

$$1 - \beta^{\text{lower}} = \delta_{\text{restricted}} + \delta_{\text{covered}} + \delta_{\text{dormant}} + \delta_{\text{residual}},$$

where the capability partition is defined by priority-filtered derived sets:

- **Restricted cell:** $R := \Xi$, where Ξ is the set of capabilities currently restricted under the preemptive-restriction criterion of [Lasser, 2026c, Proposition 1]. Classified first regardless of sacrifice evidence.
- **Covered cell:** $C := C_S \setminus \Xi$, where C_S is the set of capabilities appearing in at least one acquired bundle Y_n of a theorem-admissible sacrifice event in $[T - H, T]$. A theorem-admissible event for Part A satisfies S0, S1, S2, and S4(a); for Part B it additionally satisfies S3, S5, and within-bundle poset-independence (the event-local bundle decomposition [Lasser, 2026d]).
- **Dormant cell:** $D := D_S$, where $D_S = \{d \in P : d \notin \Xi, d \notin R_S, d \notin C_S\}$ is already priority-filtered by construction. These are tradable-in-principle capabilities with zero want-pursuit sacrifice evidence—dormant wants.
- **Residual cell:** $S := R_S \setminus \Xi$, where R_S is the structural residual class (Definition 11). Residual capabilities outside Ξ ; capabilities in $\Xi \cap R_S$ are already assigned to R by priority.

The four cells $\{R, C, D, S\}$ are pairwise disjoint and their union is P : R and S are disjoint by construction; C and D are priority-filtered to exclude Ξ ; $C \cap R_S = \emptyset$ because residual capabilities admit no sacrifice-observable path in principle (Definition 11) and therefore cannot appear in any admissible bundle; remaining pairs are disjoint by the bullet-stated exclusions.

Poset-boundary closure hypothesis. We additionally require the partition $\{R, C, D, S\}$ to be poset-boundary closed, meaning both:

- (CC1) Cooperative closure. For every cooperative capability $z \in P$, all participants of z lie in the same cell.
- (CC2) Subsumption closure. No subsumption edge $c \succeq c'$ crosses a cell boundary: if c, c' are connected by subsumption in P , they lie in the same cell.

Set-disjointness alone does not suffice for vol_P -additivity across cells; by the category partition [Lasser, 2026d](i) and the M4 axiom inherited from Proposition 1 [Lasser, 2026f], additivity requires full poset-independence—neither cross-cell cooperative composition nor cross-cell subsumption. A cooperative whose participants straddle two cells contributes to $\text{vol}_P(P)$ but to neither vol_P -cell individually (CC1 failure); a subsumption edge crossing the boundary couples vol_P -mass across cells in ways M4’s independent-component additivity cannot decompose (CC2 failure). Together (CC1) and (CC2) give the poset-independence that licenses $\text{vol}_P(P) = \sum_X \text{vol}_P(X)$ for $X \in \{R, C, D, S\}$. See Remark 10 for how to achieve poset-boundary closure in practice.

The δ terms are:

$$\begin{aligned} \delta_{\text{restricted}} &= [\text{vol}_P(R) - \text{vol}_R^{\text{lower}}(R)] / \text{vol}_P(G), \\ \delta_{\text{covered}} &= [\text{vol}_P(C) - \text{vol}_R^{\text{lower}}(C)] / \text{vol}_P(G), \\ \delta_{\text{dormant}} &= \text{vol}_P(D) / \text{vol}_P(G), \\ \delta_{\text{residual}} &= \text{vol}_P(S) / \text{vol}_P(G). \end{aligned}$$

Under poset-boundary closure, the identity $1 - \beta^{\text{lower}} = \sum \delta$ follows from $\text{vol}_P(G) = \text{vol}_P(R) + \text{vol}_P(C) + \text{vol}_P(D) + \text{vol}_P(S)$ (poset-boundary closure + set-disjoint union is P , via M4), combined

with $\text{vol}_R^{\text{lower}}(P) = \text{vol}_R^{\text{lower}}(R) + \text{vol}_R^{\text{lower}}(C) + 0 + 0$ (dormant and residual cells contribute zero $\text{vol}_R^{\text{lower}}$ by definition).

Remark 10 (Achieving poset-boundary closure of the cells). *Three operational strategies admit the poset-boundary closure hypothesis without changing the framework’s substantive content. Each strategy handles both the cooperative (CC1) and subsumption (CC2) conditions jointly, since both are instances of the “edge connects different cells” pattern; concretely, treat the poset’s combined cooperative/subsumption graph (each cooperative contributes hyperedges to its participants; each subsumption relation contributes a directed edge) as the object whose boundary-crossing instances must be eliminated.*

- (a) Priority-extension (default). *For each node $n \in P$ not yet in a settled cell, and for each subsumption edge $c \succeq c'$ or cooperative z with participant set $\Pi(z)$ touching n : take the transitive closure under cooperative participation and subsumption edges, collect the resulting component, and place the whole component in the cell of its highest-priority member (restricted > covered > dormant > residual). After one sweep, the cells are both cooperative-closed and subsumption-closed (hence poset-boundary closed), and the original individual-capability classification is preserved except where a priority-dominated node was absorbed into a higher-priority cell.*
- (b) Poset-closed sub-partition. *Refine the partition so each cell is sub-partitioned into its poset-connected components under both cooperative composition and subsumption. The gap decomposition then operates at the sub-cell level and aggregates via two M4 applications (within-cell and across-cell).*
- (c) Boundary residual. *Add a fifth bucket for boundary-crossing cooperatives and subsumption edges. The identity becomes $1 - \beta^{\text{lower}} = \delta_{\text{restricted}} + \delta_{\text{covered}} + \delta_{\text{dormant}} + \delta_{\text{residual}} + \delta_{\text{boundary}}$ with a fifth term for unassigned cross-cell poset mass (cooperative and subsumption-crossing combined).*

The paper’s stated results assume strategy (a) by default: the priority-extension rule is the natural operational choice, and the practical prevalence of boundary-crossing poset edges is small when the capability partition’s priority classes are themselves well-structured (e.g., restrictions are poset-boundary closed because restricting a cooperative implies restricting its participants, and restricting a capability implies restricting capabilities subsumed by it in cost-aligned posets).

Proposition 5 (Computability of Gap Decomposition). *Given (i) the aggregate sacrifice data (events in $[T - H, T]$), (ii) the restriction set Ξ , (iii) the residual class R_S (Definition 11), (iv) the capability census (enumeration of P) together with the poset structure $\mathcal{M}$: the cooperative-composition hypergraph (cooperative nodes and their participants) and the subsumption relation graph (directed edges $c \succeq c'$), (v) an event-admissibility labelling designating which observed events satisfy the theorem-admissibility requirements. Because Definition 7 operates under Part B (the aggregate lower bound theorem [Lasser, 2026d]), the full Part B admissibility set is required: $S_0, S_1, S_2, S_4(a), S_3, S_5$, the genuine-exchange conditions (i)–(iv) of the revealed-sacrifice event definition [Lasser, 2026d], and within-bundle poset-independence per the event-local bundle decomposition [Lasser, 2026d]; and (vi) a closure policy from Remark 10, with strategy (a) (priority-extension to cooperatives) as the default, the four-term gap decomposition is computable in time $O(|P| + E + M)$, where E is the number of observed (capability, event) membership records and $M = |\mathcal{M}|$ is the total size of the poset structure (cooperative hyperedges plus subsumption edges). The $O(|P|)$ term is per-individual-capability classification; the $O(E)$ term is one-pass aggregation over bundle-incidence records; the $O(M)$ term is the closure-policy pass, which traverses cooperative and subsumption edges jointly via (amortized) union-find. No structural counterfactual computation is required (a direct-measurement approach would require $O(|P|^2)$ counterfactual queries).*

Proof. Pass 1 (individual-capability classification, $O(|P|)$). Maintain a hash table from capabilities to their current class labels, initialized in $O(|P|)$ time. Sweep the individual capabilities in P once: for each $c \in P$, check $c \in \Xi$ (restricted table lookup, $O(1)$); if not, check $c \in R_S$ (residual classification lookup, $O(1)$); if in neither, mark as dormant. This gives an initial classification in $O(|P|)$.

Pass 2 (event-sweep, $O(E)$). Iterate through each (capability, event) incidence record. For each record (c, n) with event n admissibility-labelled satisfying the full Part B admissibility set of in-

put (v)—S0, S1, S2, S4(a), S3, S5, the genuine-exchange conditions, and within-bundle poset-independence of Y_n per the event-local bundle decomposition [Lasser, 2026d]—update c ’s class from dormant to covered (if not already restricted or residual). Events failing any component of the Part B admissibility set do not contribute coverage; in particular, events with internally cooperative bundles are filtered out by the within-bundle poset-independence check. This is $O(E)$ total work, one constant-time update per record.

Pass 3 (poset-boundary closure pass, $O(M)$). Under default strategy (a), walk the combined cooperative/subsumption graph via union-find, identifying poset-connected components: two nodes are in the same component if connected by any chain of cooperative-participation or subsumption edges. For each component, determine its highest- priority cell (restricted $>$ covered $>$ dormant $>$ residual) among member cells, and place the entire component in that cell. The cells become both cooperative-closed and subsumption-closed (hence poset-boundary closed per Definition 7) by construction. Union-find gives amortized $O(M \cdot \alpha(M))$ complexity, essentially linear in M . Under strategy (b) or (c), the pass is replaced by the corresponding sub-partition or boundary-bucket construction over the same combined graph, with the same complexity modulo constants.

Aggregation. Each class’s aggregate vol_P and $\text{vol}_R^{\text{lower}}$ is maintained incrementally across the three passes, giving the four quotients in $O(1)$ additional time. The counterfactual comparison—removing capability c and recomputing global reachability—is not performed; only membership checks, aggregation, and closure-policy lookups occur. A direct-measurement approach requiring $O(|P|^2)$ counterfactuals is avoided because covered/dormant classification here uses bundle-appearance evidence, not marginal-contribution evidence. ■

Remark 11 (Admissibility-labelling is an external input). *S0 (calibration) and the interior component of S1 are not mechanically verifiable from the observable event fields alone. Input (v) of Proposition 5 represents an external admissibility layer: events must be labelled admissible (by trust-weighted attestation, auditor sampling, or explicit agent commitment) before the decomposition can classify capabilities into “covered.” Under committed-mode operation ([Lasser, 2026d]), the labelling is carried by the commitment protocol’s proof attachment; in unrestricted mode, the framework must rely on governance processes to supply it. The $O(|P|)$ complexity is the classification cost given the labelling, not the cost of producing the labelling itself.*

Remark 12 (Part A versus Part B granularity). *Under Part A alone (category-level aggregates, S0–S2), the decomposition operates at the category level: each partition component C_j (the category partition [Lasser, 2026d]) is classified as restricted, covered, dormant, or residual. Under committed data ([Lasser, 2026d]), S2 is not directly verifiable; the committed-mode decomposition therefore assumes S2 holds for committed events. The per-capability decomposition is the refinement available when Part B assumptions hold.*

3.3 Need-sufficiency diagnostic (companion metric)

The gap decomposition above tracks want-pursuit: how much of the collective’s above-sufficiency capability-space is evidenced by voluntary sacrifice. The need-sufficiency diagnostic is the parallel metric for below-sufficiency: how efficiently does the collective satisfy needs, and is the satisfaction downstream-safe?

Definition 8 (Need-Access Cost). *The need-access cost aggregate for trade window $[T - H, T]$ is:*

$$\text{NAC}(T) = \sum_{n \in F_{S1}} \Delta \text{vol}_P(X_n),$$

where F_{S1} is the set of events failing S1 due to below-sufficiency (Proposition 3). NAC measures the total benchmarked capability surrendered for need-satisfaction—purchasing power and time spent reaching baseline functionality rather than pursuing wants.

NAC does not enter the aggregate lower bound theorem [Lasser, 2026d]’s aggregate lower bound or β^{lower} . It is a separate diagnostic that tracks the cost side of the agent’s capability budget.

Remark 13 (NAC is an upper bound on need-access cost). *Because the whole-event filter sends the entire $\Delta \text{vol}_P(X_n)$ of mixed need/want events into NAC, NAC overestimates the true need-access cost by including any attached want-pursuit spend. NAC should be read as an upper bound on need-access cost, not a precise measurement.*

Definition 9 (Need-Satisfaction Efficiency). *For each need N identified in the poset (Definition 1) and each agent i , the per-agent need-satisfaction efficiency is a triple:*

$$\text{NSE}(i, N) = (\text{sufficiency_status}, \text{downstream_safety}, \text{access_cost}),$$

where:

- $\text{sufficiency_status} \in \{\text{below}, \text{met}, \text{exceeded}\};$
- $\text{downstream_safety} \in \{\text{safe}, \text{degrading}, \text{unknown}\};$
- $\text{access_cost} = \sum \Delta \text{vol}_P(X_n)$ for below-sufficiency events by agent i targeting need N .

The collective-level aggregate $\text{NSE}(T)$ reports the distribution of (agent, need) pairs across sufficiency statuses and flags degrading pathways.

Remark 14 (Observability under commitment). *When sacrifice events are committed ([Lasser, 2026d]), per-agent $\text{NSE}(i, N)$ is not recoverable. The committed-mode fallback is a population-level proxy: $\text{NSE}_{\text{proxy}}(T)$ reports $\text{NAC}(T)$ by need category and counts of below-sufficiency events. Full $\text{NSE}(T)$ requires agents to voluntarily attest pathways or the framework to operate in non-committed mode.*

Proposition 6 (Combined Diagnostic). *The framework reports the triple $(\text{vol}_R^{\text{lower}}, \text{NAC}, \text{NSE})$ as a joint diagnostic:*

- (a) **High $\text{vol}_R^{\text{lower}}$ + low NAC:** *Healthy. The collective exercises rich capabilities above sufficiency and needs are cheap to meet.*
- (b) **Low $\text{vol}_R^{\text{lower}}$ + high NAC:** *Proxy failure on needs. vol_P may be high, but agents are spending most of their benchmarked capability budget on basic need-satisfaction. The sacrifice channel correctly excludes these events (S1 fails), so the low $\text{vol}_R^{\text{lower}}$ accurately reflects low above-sufficiency exercise. NAC surfaces why $\text{vol}_R^{\text{lower}}$ is low.*
- (c) **Low $\text{vol}_R^{\text{lower}}$ + low NAC:** *Dormancy, residual, or theorem-inadmissible activity. The gap decomposition (Definition 7) distinguishes dormant from residual.*
- (d) **High $\text{vol}_R^{\text{lower}}$ + high NAC:** *Mixed regime. NSE distinguishes “expensive but safe” from “expensive and degrading.”*

Remark 15 (Parallel structure, not nesting). *The need-sufficiency diagnostic is parallel to the gap decomposition, not nested within it, but operates on different formal objects. The gap decomposition is a capability partition; NAC is an event aggregate; NSE is a per-need state assessment. The two diagnostics cover complementary portions of the sacrifice event stream: S1-satisfying events (that also satisfy S0 and S2) enter the gap decomposition via the aggregate lower bound theorem [Lasser, 2026d]; events failing S1 due to below-sufficiency enter NAC. Events failing S1 for non-need reasons (coercion, duress) are excluded from both diagnostics and constitute an undiagnosed event-stream residual.*

4 Wireheading-Consistent Signature via Trade-Flow Concentration

Wireheading produces high β^{lower} with degenerate exercise: the collective exercises capabilities, but only in narrow self-reinforcing loops. Under the sacrifice-based observation model, the wireheading signature appears as *trade-flow concentration*: the collective’s sacrifice events cluster in a narrow category of unbenchmarked bundles, rather than distributing across the capability space.

Definition 10 (Trade-Flow Leverage). *For bundle category $C_j \in \mathcal{C}$ (the category partition [Lasser, 2026d]), the trade-flow leverage of C_j at time T is:*

$$\lambda_{\text{trade}}(j, T) = \begin{cases} \text{vol}_R^{\text{lower}}(C_j, T) / \text{vol}_R^{\text{lower}}(G, T) & \text{if } \text{vol}_R^{\text{lower}}(G, T) > 0, \\ \text{undefined (diagnostic suppressed)} & \text{if } \text{vol}_R^{\text{lower}}(G, T) = 0. \end{cases}$$

Under healthy trade diversity, leverage is distributed across many categories. Under wireheading, leverage concentrates on the loop drivers. When total observed trade flow is zero (the agent has produced no theorem-admissible sacrifice events in the window), λ_{trade} is not a meaningful quantity and the diagnostic is suppressed; the relevant signal in that regime is the gap decomposition (§3), which classifies capabilities as dormant or residual rather than as concentration-anomalies.

Proposition 7 (Wireheading-Consistent Concentration Signal). *When $\text{vol}_R^{\text{lower}}(G, T) > 0$, the Herfindahl–Hirschman Index [Rhoades, 1993]*

$$\text{HHI}(\lambda_{\text{trade}}) = \sum_j \lambda_{\text{trade}}(j)^2$$

is Schur-convex on the simplex of probability vectors: for leverage distributions λ, λ' over the same support, if λ majorizes λ' (i.e., λ is at least as concentrated as λ' in the sense of Hardy et al. [1929]), then $\text{HHI}(\lambda) \geq \text{HHI}(\lambda')$. Concentration in the majorization order therefore induces a monotone ordering of HHI values, and comparing HHI against a governance-set threshold $\text{HHI}_{\text{alarm}}$ provides a concentration-consistent-with-wireheading alarm: if $\text{HHI} > \text{HHI}_{\text{alarm}}$, a small number of bundle categories dominate the aggregate, which is the behavioural signature of narrow self-reinforcing exercise.

Remark 16 ($\text{HHI}_{\text{alarm}}$ is a governance parameter, not a theorem-derived separator). *The alarm threshold $\text{HHI}_{\text{alarm}}$ is not derived from the framework’s axioms; it is a governance parameter calibrated to the framework’s tolerance for concentration risk, analogous to HHI thresholds in antitrust practice and to the sufficiency thresholds of Definition 1. Its appropriate value depends on the support size (number of categories with non-zero leverage), the domain’s expected baseline concentration, and the framework’s false-positive/false-negative tolerance. The structural guarantee is that HHI is monotone in concentration; the separation between wireheading-consistent and wireheading-inconsistent patterns is operational, not theorem-derived.*

Proof. Let $m = |\{j : \text{vol}_R^{\text{lower}}(C_j) > 0\}|$ be the number of bundle categories with non-zero observed leverage. The leverage vector $\lambda = (\lambda_{\text{trade}}(j))_j$ is a probability distribution: $\lambda_j \geq 0$ and $\sum_j \lambda_j = 1$ by the normalization in Definition 10.

Schur-convexity. $\text{HHI}(\lambda) = \sum_j \lambda_j^2$ is a sum of convex functions of the coordinates and is symmetric under permutation. Any symmetric convex function on the simplex is Schur-convex by the Hardy–Littlewood–Pólya theorem [Hardy et al., 1929, Marshall et al., 2011]; therefore λ majorizing λ' implies $\text{HHI}(\lambda) \geq \text{HHI}(\lambda')$. Equivalently, spreading mass from heavier bins to lighter bins weakly decreases HHI, and concentrating mass from lighter bins to heavier bins weakly increases it.

Quantitative bounds. For any probability distribution λ with support of size m , Cauchy–Schwarz (or Jensen’s inequality on the squaring map) gives $\text{HHI}(\lambda) \geq 1/m$, with equality iff λ is uniform—the minimally-concentrated distribution in the majorization order. At the other extreme, if the top k categories capture fraction ρ of total leverage, $\text{HHI} \geq \rho^2/k$ (achieved by concentrating the top- k mass uniformly on those k bins).

Alarm criterion. A wireheading signature concentrates trade flow in k loop-driver categories with ρ close to 1 and k small, forcing $\text{HHI} \geq \rho^2/k$. A governance-chosen threshold $\text{HHI}_{\text{alarm}} \in (1/m, 1)$ determines which concentration levels trigger the alarm; by Schur-convexity, the alarm fires on a majorization-upper set of distributions (if λ triggers the alarm and $\lambda' \succeq \lambda$ in the majorization order, then λ' also triggers it). Whether any specific $\text{HHI}_{\text{alarm}}$ cleanly separates “diverse” from “loop-dominated” depends on the domain’s baseline concentration and the framework’s tolerance for false positives/negatives—the threshold is a governance parameter, not a theorem-derived separator (see Remark 16). ■

Remark 17 (Advantage over structural counterfactual). *A direct-measurement approach would try to identify wireheading via exercise-leverage concentration, requiring the marginal vol_R contribution of each capability (a counterfactual query). The trade-flow formulation surfaces the concentration signal via HHI on observed trade data, requiring no counterfactual computation. The diagnostic is strictly more tractable; it remains a signal rather than a definitive detection, per Remark 16.*

Proposition 8 (Third-Party Observability of the Concentration Signal (Part A)). *When the total observed sacrifice magnitude $\sum_n \Delta \text{vol}_P(X_n) > 0$, the Part A category-flow concentration statistic $\widetilde{\text{HHI}}$ is computable from public trade data (committed sacrifice events, the committed event definition [Lasser, 2026d]) without access to any agent’s private state. Specifically, define*

$$\tilde{\lambda}_{\text{trade}}(j, T) = \begin{cases} \frac{\sum_{n: \text{category}(Y)_{[n]}=j} \Delta \text{vol}_P(X_n)}{\sum_n \Delta \text{vol}_P(X_n)} & \text{if } \sum_n \Delta \text{vol}_P(X_n) > 0, \\ \text{undefined (diagnostic suppressed)} & \text{if } \sum_n \Delta \text{vol}_P(X_n) = 0, \end{cases} \quad \widetilde{\text{HHI}} = \sum_j \tilde{\lambda}_{\text{trade}}(j)^2,$$

for the trade window T (with $\widetilde{\text{HHI}}$ correspondingly suppressed when $\tilde{\lambda}_{\text{trade}}$ is undefined). A third-party auditor with access only to the committed sacrifice ledger can compute $\widetilde{\text{HHI}}$ whenever the window contains positive-magnitude events and raise the Part A category-flow alarm (wireheading-consistent in the sense of Proposition 7, with $\tilde{\lambda}_{\text{trade}}$ substituted for λ_{trade} ; not a wireheading determination on its own). When the window has zero total sacrifice flow, the diagnostic is suppressed (the relevant signal in that regime is the gap decomposition, §3, matching the parallel suppression rule of Definition 10).

Proof. By the committed event definition [Lasser, 2026d], every committed sacrifice event exposes the pair $(\text{category}(Y)[n], \Delta\text{vol}_P(X_n))$ in cleartext. For the category-flow statistic, both the numerator $\sum_{n: \text{category}(Y)[n]=j} \Delta\text{vol}_P(X_n)$ (which sums only committed fields conditional on the committed category label) and the denominator $\sum_n \Delta\text{vol}_P(X_n)$ are direct functions of the revealed-cleartext fields. No hidden field—agent identity, bundle-interior subset, $\alpha_{n,c}$ coefficients—enters the computation. Hence $\widetilde{\text{HHI}} = \sum_j \tilde{\lambda}_{\text{trade}}(j)^2$ is publicly computable. The alarm test $\widetilde{\text{HHI}} > \text{HHI}_{\text{alarm}}$ is consequently computable by any party with read access to the committed-event ledger. Monotonicity of $\widetilde{\text{HHI}}$ in trade-flow concentration across categories follows from the same Cauchy–Schwarz / Herfindahl argument as Proposition 7, applied to the category-flow distribution. ■

Remark 18 (Part B observability requires richer commitments). *The per-capability Part B concentration statistic $\text{HHI}(\lambda_{\text{trade}})$ built on $\text{vol}_R^{\text{lower}}(C_j, T) = \sum_{c \in C_j \cap U} \text{vol}_R^{\text{lower}}(c, T)$ is not third-party computable from the committed-event interface of the committed event definition [Lasser, 2026d]. The committed form reveals only the partition label $\text{category}(Y)[n]$ and attests $Y_n \subseteq C_{\text{category}(Y)[n]}$; it does not reveal which subset of $C_{\text{category}(Y)[n]}$ was actually acquired, nor the per-capability attribution coefficients $\alpha_{n,c}$. Computing $\text{vol}_R^{\text{lower}}(c, T) = \max_{n: c \in Y_n} \alpha_{n,c} \cdot \Delta\text{vol}_P(X_n)$ therefore requires additional committed fields: per-event subset-membership openings (or a ZK attestation equivalent), and the $\alpha_{n,c}$ coefficients (public by an equal-weight default, or committed alongside the event under a richer Part B variant as flagged in the Part B commitment remark [Lasser, 2026d]). Under that richer commitment, the Part B statistic becomes third-party observable; under the default Part A commitment, it is not. Within a consented operational perimeter O (the exercise-indicator appendix of [Lasser, 2026d]), no richer commitment is needed because direct per-capability evaluation is permitted; the Part B statistic is then available to the perimeter operator, not to arbitrary third parties.*

5 Residual Class Under Revealed-Sacrifice Observation

5.1 Re-characterization

A natural alternative approach would define the residual class as capabilities failing benchmarkability (identity-individuated, non-repeatable, non-communicable). The revealed-sacrifice framing produces a differently-shaped and operationally cleaner residual class.

Definition 11 (Residual Class Under Revealed Sacrifice). *The residual class is the set of capabilities structurally outside the reach of the sacrifice channel—capabilities whose exercise, by their nature, produces no sacrifice event even with unlimited observation time:*

$$R_S = \{c \in P : \text{no sacrifice-observable path exists in principle}\}.$$

R_S is a structural property of the capability, not a property of the observation window. A capability is in R_S because the sacrifice channel cannot reach it (purely private exercise with no material footprint), not because the channel has not yet observed it. Examples: private contemplative experience, purely internal cognitive operations with no external inputs or time diversion.

Remark 19 (Structural versus evidential absence). R_S is distinct from the set of capabilities with no sacrifice evidence in a given window. A capability outside R_S (tradable in principle) may have no sacrifice evidence in a particular window simply because no trade happened to cover it—this is the dormant category in the gap decomposition (Definition 7), not the residual class. Dormant capabilities can be resolved by waiting for more trade data; residual capabilities cannot.

Remark 20 (Conservatism of R_S classification). *Classifying a capability as residual is a structural claim that should be conservative: if there is any plausible mechanism by which exercise of c could produce a sacrifice event (even indirectly—e.g., an artistic practice that requires purchasing supplies), c should be classified as outside R_S . The residual class should be small by construction.*

5.2 Relationship to the benchmarkability-based residual class

Proposition 9 (Residual Class Intersection). *Let R_B be the benchmarkability-based residual class (capabilities failing benchmarkability under a direct-measurement approach) and R_S be the revealed-sacrifice residual class (Definition 11). Then:*

- (a) R_B and R_S overlap but neither contains the other.
- (b) Capabilities in $R_B \setminus R_S$: *identity-individuated capabilities that nonetheless involve observable sacrifice. Example: an artistic practice that is identity-individuated but involves purchasing supplies, paying for studio space, foregoing labour hours. The time-sacrifice channel captures a lower bound even though the capability is identity-individuated.*
- (c) Capabilities in $R_S \setminus R_B$: *capabilities that are benchmarkable (communicable, repeatable, externally verifiable) but are exercised purely internally with no trade event. Example: a cognitive capability (mathematical reasoning, strategic planning) that the agent exercises entirely in private thought.*
- (d) Capabilities in $R_B \cap R_S$: *identity-individuated capabilities exercised purely privately with no sacrifice event. Example: private contemplative experience with no material input or time-sacrifice signal.*

Proof. Each part is established by exhibiting a capability with the claimed membership, which proves non-containment in the relevant direction.

(a) *Neither class contains the other.* Parts (b) and (c) below produce witnesses for $R_B \setminus R_S \neq \emptyset$ and $R_S \setminus R_B \neq \emptyset$ respectively, ruling out either containment.

(b) $R_B \setminus R_S$. Consider the capability “a specific artist’s studio practice.” Under the benchmarkability criterion of [Lasser, 2026f, Definition 2], this is identity-individuated: the artefact sequence is non-repeatable and non-communicable in the benchmark sense, so the capability lies in R_B . But the practice consumes materials (paid-for supplies, rent on studio space) and displaces outside-option labour hours; every session produces a money- or time-sacrifice event ([Lasser, 2026d]) that emits a valid $\text{vol}_R^{\text{lower}}$ signal on the underlying bundle. Thus the capability is reachable in principle by the sacrifice channel and so $\notin R_S$.

(c) $R_S \setminus R_B$. The residual definition (Definition 11) requires that *no* exercise mode produces a sacrifice event in principle—not merely that some modes do not. Many cognitive capabilities that feel purely internal (silent rehearsal, planning, reasoning) admit paid-training or dedicated-time exercise modes and therefore lie outside R_S . The genuine witnesses for $R_S \setminus R_B$ are capabilities exercised passively and concurrently with other activity, with no incremental material input, no incremental time displacement, and no counterparty.

Consider “ongoing interoceptive self-awareness (the felt sense of one’s internal physiological state while engaged in other activity).” This capability is benchmarkable: interoceptive acuity is measurable in clinical neuroscience, communicable through structured report, and repeatable across sessions, so it lies outside R_B . But interoception is exercised as a background condition of conscious embodiment, concurrent with every other activity the agent performs; no payment secures it independently of the body, and no hour of foregone wages is *specifically* attributable to it (the same hours simultaneously produce whatever else the agent is doing). No sacrifice event, in principle, distinguishes the incremental exercise of this capability from the baseline of being conscious. Under the conservative reading of R_S (Remark 20), this is a genuine witness. The same structural argument applies to peripheral sensory integration, background attentional stability, and other passively-concurrent capabilities.

(d) $R_B \cap R_S$. Consider “private contemplative experience with no material footprint.” This is identity-individuated (non-communicable phenomenology, so $\in R_B$) and emits no sacrifice event

(no money, no time beyond idle cycles, no counterparty, so $\in R_S$). The intersection is witnessed and therefore non-empty. ■

Remark 21 (The new residual class is the correct curve). *The prior residual class was defined by the measurement apparatus (what benchmarks can test). The new residual class is defined by the observation channel (what trades reveal). The observation-channel curve is the correct one for a paper about what the framework can observe under its privacy commitments: the question is not “what could we test if we had access?” but “what do we actually see?”*

5.3 Bounding the residual class

Paper 3’s observational-individuation result [Lasser, 2026c, Corollary 2.1] establishes a *structural* floor on vol_P : for any distinguishable agent k , the poset contains an identity capability d_k with $\text{OI}_{\text{floor}}(k) \geq 1$, so removing a_k contracts vol_P by at least $\text{OI}_{\text{floor}}(k)$. Paper 8’s $\text{vol}_R^{\text{exact}}(O)$ diagnostic (the exercise indicator [Lasser, 2026d]), by contrast, sums vol_P -weights of capabilities d with $e_t(d) = 1$, where $e_t(d)$ requires a realized cooperative output in $[t - \Delta, t]$ that would be unrealizable if d were removed from O . Transferring the OI floor into $\text{vol}_R^{\text{exact}}(O)$ therefore requires a bridge between the structural vol_P fact and the temporal exercise indicator. We state that bridge explicitly as a hypothesis rather than derive it from Corollary 2.1.

Definition 12 (OI–Exercise Bridge Condition). *An agent k with OI capability $d_k \in O$ satisfies the OI–exercise bridge at time t if some realized cooperative output in $[t - \Delta, t]$ that agent k participated in is identity-individuating in the sense that removing d_k from O would make that output unrealizable as an output attributable to k . Equivalently, $e_t(d_k) = 1$ under the exercise indicator [Lasser, 2026d]. Active presence within O that is merely dormant (e.g., k is nominally in the perimeter but contributed nothing to realized cooperative outputs during $[t - \Delta, t]$) does not satisfy the bridge.*

Proposition 10 (Active-Agent Floor Transfer, conditional on the bridge). *Fix a window $W = [t_0, t_0 + T]$. For an agent k with OI capability $d_k \in O$ satisfying the OI–exercise bridge (Definition 12) at some $t \in W$, d_k is exercised within W and therefore contributes $\text{OI}_{\text{floor}}(k) \geq 1$ to $\text{vol}_R^{\text{exact}[W]}(O)$ (with the weight bound from the binary default of [Lasser, 2026c, Corollary 2.1]). Summing over agents $k \in K$ whose OI capabilities are pairwise poset-independent (the category partition [Lasser, 2026d](i)) and who each satisfy the bridge at some $t \in W$:*

$$\text{vol}_R^{\text{exact}[W]}(O) \geq \sum_{k \in K} \text{OI}_{\text{floor}}(k) =: \sum_{k \in K} \text{OI}_{\text{floor}}(k).$$

The floor transfers only under the bridge hypothesis; it is not a consequence of Corollary 2.1 alone, which establishes a vol_P elimination floor rather than an exercise witness.

Proof. By the bridge (Definition 12) at some $t \in W$, $e_t(d_k) = 1$, so $d_k \in P_t^{\text{ex}}$ for that t . Under Paper 8’s window-integrated form (the vol_R definition [Lasser, 2026d]), $d_k \in P_W^{\text{ex}} = \bigcup_{s \in W} P_s^{\text{ex}}$, hence d_k contributes $\text{OI}_{\text{floor}}(k)$ to $\text{vol}_R^{\text{exact}[W]}(O) = \text{vol}_P(O \cap P_W^{\text{ex}})$. By Paper 3 [Lasser, 2026c, Corollary 2.1], $\text{OI}_{\text{floor}}(k) \geq 1$ (the binary default, strengthened by any subsumption weight in vol_P). Pairwise poset-independence of the $\{d_k\}_{k \in K}$ lets vol_P decompose additively by axiom M4 ([Lasser, 2026d]), giving the sum form. Without the bridge hypothesis, Corollary 2.1 [Lasser, 2026c] establishes only the structural vol_P -on-elimination fact, which does *not* supply the exercise witness required by the exercise indicator [Lasser, 2026d]. ■

Remark 22 (When the bridge holds and when it fails). *The bridge holds when active participation is identity-coupled: the agent’s actions produce realized cooperative outputs whose attribution depends on the agent remaining in the perimeter. Typical operational examples are participation in a committed trade event, execution of a verified action with agent-attributable signing, or contribution to a cooperative capability whose output is signed by the agent’s identifier. The bridge fails for dormant presence: an agent is nominally within O but contributed nothing attributable to them in $[t - \Delta, t]$. In such windows the structural vol_P floor of Corollary 2.1 still holds (removing the agent would still contract vol_P), but $\text{vol}_R^{\text{exact}}(O)$ receives no contribution from d_k until the agent resumes identity-coupled activity. This is consistent with the paper’s stance that vol_R measures what the collective does, not merely what it possesses.*

Remark 23 (Channel separation). *Under the boundary-closed hypothesis on O (the exercise indicator [Lasser, 2026d], conditions BC1/BC2), the OI floor contributes exclusively to $\text{vol}_R^{\text{exact}}(O)$ in the combined diagnostic (the exercise-indicator appendix of [Lasser, 2026d]):*

$$\text{vol}_R^{\text{combined}} = \text{vol}_R^{\text{exact}}(O) + \text{vol}_R^{\text{lower}}(P \setminus O).$$

It does not contribute to $\text{vol}_R^{\text{lower}}$ (the sacrifice-based bound outside O). Conflating the two channels would double-count. When boundary-closure fails (either BC1 or BC2), $\text{vol}_R^{\text{combined}}$ remains a conservative lower bound on $\text{vol}_R(P)$ but undercounts the cross-boundary cooperatives and subsumption mass (see the boundary-cooperative-undercount remark in [Lasser, 2026d]); the OI floor’s contribution is unaffected because the OI capability d_k is, by construction, an internal capability of agent k ’s substrate—not a cooperative or subsumption edge spanning the perimeter boundary.

6 Worked Example: The Dormant-Capability Scenario Re-Staged

6.1 Setup

Two-substrate collective (biology + silicon):

- 8 individual capabilities: 3 biological (b_1, b_2, b_3), 5 silicon ($s_1, \dots, s_5$).
- 8 cooperative capabilities: $\{b_1, s_1\}, \{b_1, s_2\}, \{b_2, s_3\}, \{b_2, s_4\}, \{b_3, s_5\}, \{s_1, s_2\}, \{b_1, b_2\}, \{s_3, s_4\}$.
- 3 private contemplative capabilities (p_1, p_2, p_3) in the residual class R_S —purely private exercise, no sacrifice-observable path.
- Weights: $w(b_i) = 2.0, w(s_j) = 1.0, w(\text{cooperative}) = 1.0, w(p_k) = 1.0$.
- Total $\text{vol}_P = 3 \times 2 + 5 \times 1 + 8 \times 1 + 3 \times 1 = 22.0$.

6.2 Phase 1: Healthy trade (β^{lower} near 1)

The collective operates with diverse trade patterns. Biological agents sacrifice purchasing power and time for bundles that exercise biological capabilities: creative supplies, relationship-building time, embodied experiences. Silicon agents sacrifice compute resources for unbenchmarked optimization targets.

Trade data. 50 above-sufficiency sacrifice events over the window, covering all 16 non-residual capabilities. Additionally, 12 below-sufficiency events are observed (biological agents purchasing healthcare, food, shelter) but these fail S1 (Proposition 3) and do not enter the sacrifice aggregate.

Diagnostics. $\beta^{\text{lower}} = 19.0/22.0 = 0.864$. Gap decomposition: $\delta_{\text{restricted}} = 0, \delta_{\text{covered}} = 0, \delta_{\text{dormant}} = 0, \delta_{\text{residual}} = 3.0/22.0 = 0.136$ (residual class contribution). No alarm.

Need-sufficiency diagnostic. $\text{NAC} = 3.5$ (an upper bound on true need-access costs per Remark 13). NSE : all biological needs at $\text{sufficiency_status} = \text{met}$, $\text{downstream_safety} = \text{safe}$. Combined diagnostic: ($\text{vol}_R^{\text{lower}} = 19.0, \text{NAC} = 3.5, \text{NSE} = \text{healthy}$). The collective exercises rich above-sufficiency capabilities and needs are cheap to meet.

6.3 Phase 2: Automation concentrates trade (β^{lower} drops)

Silicon capabilities automate tasks previously requiring biological participation. The automation changes *trade patterns*: the biological agents no longer sacrifice time for activities involving b_2 and b_3 .

Trade data. Trade events for b_2 -related bundles: $8 \rightarrow 0$. Trade events for b_3 -related bundles: $6 \rightarrow 0$. Silicon trade events increase. New aggregate: 52 above-sufficiency events covering only 10 of 16 non-residual capabilities.

Diagnostics. $\beta^{\text{lower}} = 10.0/22.0 = 0.455$. **ALARM fires** ($\beta^{\text{lower}} < 0.8$). Gap decomposition:

$$\begin{aligned}\delta_{\text{restricted}} &= 0, \\ \delta_{\text{covered}} &= 1.0/22 = 0.045, \\ \delta_{\text{dormant}} &= 8.0/22 = 0.364 \quad (b_2, b_3, \text{ and } 4 \text{ cooperatives: dormant wants}), \\ \delta_{\text{residual}} &= 3.0/22 = 0.136.\end{aligned}$$

Check:

$$\beta^{\text{lower}} + \delta_{\text{covered}} + \delta_{\text{dormant}} + \delta_{\text{residual}} = 0.455 + 0.045 + 0.364 + 0.136 = 1.000.$$

Need-sufficiency diagnostic. NAC rises from 3.5 to 8.2 as automation displaces biological agents from cooperative outputs, forcing them to spend more on basic need-satisfaction (healthcare costs rise without cooperative support structures). NSE: healthcare need for biological agents shifts from downstream_safety = safe to downstream_safety = degrading (cooperative pathways that previously contributed to healthcare quality are no longer active).

Combined diagnostic: ($\text{vol}_R^{\text{lower}} = 10.0$, NAC = 8.2, NSE = degrading on healthcare). The need-sufficiency diagnostic catches a failure mode the gap decomposition alone would miss: automation is not merely making want-pursuit capabilities dormant—it is also degrading the need-satisfaction pathways for biological agents.

6.4 Phase 3: Diagnosis and recovery

The framework diagnoses: two biological capabilities have zero sacrifice evidence. The alarm identifies the *absence of trade signal for biological-valued bundles* as the cause.

Recovery mechanism. The framework does not *impose* exercise (that would be controlled relaxation [Lasser, 2026h], for the risk dimension). Instead, the framework surfaces the diagnostic to the collective: “biological capabilities b_2 and b_3 have zero sacrifice evidence; the collective’s vol_P includes these capabilities but no agent is trading for them.” The collective can then:

- (a) Investigate whether the capabilities are genuinely dormant (no one values them anymore) or merely displaced (valued but crowded out by automation).
- (b) If displaced: restructure cooperative arrangements to create trade opportunities.
- (c) If genuinely dormant: accept the lower β^{lower} and potentially revise downward the capabilities’ contribution to vol_P .

6.5 Phase 4: The complementary diagnostic

Controlled relaxation [Lasser, 2026h] asks: are the restrictions on biological capabilities warranted? The revealed-sacrifice channel asks: are the biological capabilities worth recovering?

If the biological capabilities are under restriction, controlled relaxation tests whether the risk is real. If they are merely dormant (not restricted), the sacrifice channel detects the absence and surfaces it. The two papers cover both failure modes: over-restriction (Lasser, 2026h) and under-exercise (the present paper).

6.6 Lessons from the example

1. **Automation changes trade patterns, not just exercise patterns.** The sacrifice-based diagnostic detects dormancy through the *absence* of trade signal, not through a counterfactual computation.
2. **The alarm detects the problem without interior access.** The framework observes aggregate trade data, not individual agent behaviour.
3. **Recovery is cooperative, not coercive.** The framework surfaces the diagnostic; the collective decides what to do.

4. **The dynamics parallel the phase-boundary analysis of [Lasser, 2026e].** Dormancy of biological capabilities in the trade data is the same underlying phenomenon as observation-channel degradation: biological contribution to the collective decreases. The phase-boundary analysis treats the dynamical consequence (world-model degradation); the present paper treats the value consequence (vol_P no longer tracks exercised optionality).

7 Integration: Architecture Complements the Measurement Theorem

7.1 Position in the GFM sequence

The companion paper [Lasser, 2026d] contributes the measurement theorem—a constructive lower bound on vol_R from voluntary trade events. This paper contributes the architectural layers that make that theorem *operational*: the filter that keeps its admissibility hypotheses satisfied (need-sufficiency), the decomposition that interprets its output (δ -partition of the gap), the concentration-based diagnostic that catches one specific failure mode (wireheading HHI), and the characterization of what the measurement cannot reach (residual class).

The pair fits the preemptive-restriction criterion [Lasser, 2026c, Proposition 1] as follows. Controlled relaxation [Lasser, 2026h] tests the risk dimension; revealed sacrifice (companion paper) tests the value dimension. When a restriction decision is pending, controlled relaxation asks *is the feared harm real?* and the measurement theorem asks *is the forgone capability valuable enough to recover?* The present paper adds: *how should we interpret the measurement when it fires an alarm, and what is it structurally unable to see?*

7.2 Connection to the Goodhart mechanism

The need-sufficiency architecture (Section 2) gives a concrete mechanism for the kind of proxy failure studied in the Goodhart literature [Strathern, 1997, Manheim and Garrabrant, 2019]. A framework that fails to distinguish needs from wants—or fails to invert cost-dimension benchmarks—produces a proxy vol_P that is systematically misaligned with vol_R on need dimensions: rewarding agents for having expensive-to-satisfy needs (high cost = high sacrifice = high “revealed value” under a naive reading) when the correct interpretation is the opposite.

The polarity boundary (Definition 3) is the framework’s prophylactic against this specific Goodhart mechanism. Establishing the boundary and filtering below-sufficiency sacrifices via $S1$ *before* the sacrifice channel operates prevents need-cost inflation from corrupting the vol_R lower bound. The NAC diagnostic (Definition 8) tracks the excluded need-costs separately, making the Goodhart pressure visible rather than hidden.

7.3 Connection to the capability approach

The below-sufficiency / above-sufficiency partition recovers the distinction between basic capabilities and complex capabilities of Sen [1999] and Nussbaum [2011] within the single-poset treatment. The framework adds three formal instruments the philosophical tradition lacks: (i) the downstream-safety condition (Definition 2), which makes “sufficient” structurally rigorous rather than normatively asserted; (ii) the polarity-correct benchmark discipline (Definition 4), which aligns vol_P -maximization with the agent’s interest on both sides of the threshold; and (iii) the revealed-sacrifice transition (mechanism (c), Section 2.5), which identifies the sufficiency threshold empirically.

A fuller treatment of the correspondence between GFM and the Sen–Nussbaum capability approach, including a formal fungibility-collapse theorem reducing GFM to standard microeconomics as a special case, is the subject of a companion paper in preparation.

7.4 The shared privacy discipline

The present paper inherits the privacy discipline established in the companion paper and the broader GFM sequence: *resolution through frequency, not through depth*. The gap-decomposition computability (Proposition 5) operates in $O(|P| + E + M)$ using only public inputs (capability census, poset structure, admissibility labels, closure policy); the wireheading-consistent signal (Proposition 8) is third-party observable from public committed trade data. Neither diagnostic requires

deeper access to individual agent state than the companion paper’s measurement channel already permits.

8 Discussion and Open Questions

8.1 What this paper establishes

Need-sufficiency resolves the polarity problem. The sufficiency threshold identifies the structural boundary where sacrifice polarity flips. Below-sufficiency sacrifices are involuntary and correctly filtered by S1 (Proposition 3). The need-access cost diagnostic (Definitions 8–9) tracks need-satisfaction efficiency as a companion metric.

The B-to-C gap decomposes into disjoint diagnostic cells. Under poset-boundary closure, $1 - \beta^{\text{lower}} = \delta_{\text{restricted}} + \delta_{\text{covered}} + \delta_{\text{dormant}} + \delta_{\text{residual}}$ (Definition 7). The partition is computable in $O(|P| + E + M)$ (Proposition 5) given the capability census, poset structure, restriction and residual sets, event admissibility labelling, and a closure policy. Alarm firings can be interpreted diagnostically by which δ dominates.

Wireheading is detectable from trade data. Trade-flow concentration (Proposition 7) provides a tractable wireheading-consistent concentration signal with Schur-convexity as its formal monotonicity basis. The Part-A category-flow variant is third-party observable from public committed sacrifice events (Proposition 8).

The residual class is structurally irreducible. Capabilities in R_S (Definition 11) have zero sacrifice evidence by construction and cannot gain it regardless of observation time. The residual class is the irreducible component of the gap decomposition. The revealed-observation curve of R_S (as the set of capabilities *structurally outside the channel’s reach*) differs from and is operationally cleaner than the classical benchmarkability-based curve.

8.2 Limitations

Threshold governance is not framework-derived. The sufficiency thresholds s_k and the admissible reference set Π_N^{ref} (Definition 2) are governance inputs, not framework-computed. The framework reads them at evaluation time and does not attempt to derive them from first principles.

The filter’s interaction with capability-stuffing. Proposition 4 establishes a pointwise below-sufficiency cap. This is *not* an aggregate dimension-level cap: distinct capabilities each satisfying sufficiency on the same dimension each score 1 and accumulate under M5. Full structural resolution of capability-stuffing requires machinery outside this paper’s scope.

The gap decomposition requires poset-boundary closure. The exact identity $1 - \beta^{\text{lower}} = \sum \delta$ holds only when the cells $\{R, C, D, S\}$ are poset-boundary closed (CC1 + CC2 in Definition 7). Remark 10 describes three strategies to achieve closure in practice.

The residual class is characterized but not valued. The residual class (Section 5) characterizes *which* capabilities the framework cannot reach, not *how* to value them.

8.3 Open questions

1. **Threshold governance.** Several interrelated questions surround the sufficiency thresholds. (a) *Sensitivity*: if the threshold is set too low, need-costs between the threshold and true sufficiency are misclassified as want-pursuit events; if too high, the framework over-constrains the need. (b) *Dynamic update*: thresholds may shift as technology and social context evolve; a mechanism for threshold revision (analogous to controlled relaxation applied to sufficiency standards) is needed. (c) *Cross-cultural variation*: different populations may set thresholds differently; can the framework maintain culturally-indexed thresholds while preserving a coherent aggregate diagnostic, or does variation fragment it?

2. **Need identification and M5 interaction.** How does the framework identify which capabilities are needs versus wants? The poset-structural approach (high fan-out = need) is a heuristic, not infallible; a hybrid of structural position and revealed-sacrifice polarity transition is likely required. Relatedly, under the capped-benchmark discipline (Definition 4), sufficiency-met needs contribute a fixed cap whose weight is a design parameter affecting the relative importance of need-satisfaction versus want-pursuit in vol_P .

3. **Capability-stuffing as structural avoidance.** Capability-stuffing (Section 2.4) remains an open M5 pathology under this paper’s machinery. No signature over post-declaration poset structure or exercise pattern separates stuffed from legitimate capabilities in a way the agent cannot adversarially manipulate. The canonical example reveals itself as an instance of *structural avoidance* flagged as an open gap in [Lasser, 2026f, Discussion]: the agent declines to discover that “travel from home to gas station” and “travel from work to gas station” admit merger into a canonical multi-origin fuel-access capability. Related avoidance pathologies include provenance-free declaration, the Doll Problem (declining exercise), and wireheading (declining broad engagement); these share the structural-avoidance shape but differ in formal detail.

The intended resolution is a *proof-burden taxonomy* with claim-and-challenge back-pressure. Three burden types: proof-of-non-mergeability (stuffing-specific), proof-of-provenance (blocking pure-label declarations), and proof-of-use (blocking persistently-dormant declarations outside the residual class). Proof-burdens operate as canonicalization preprocessing on P before M5 is applied; M5’s formula is preserved, but the preprocessing introduces substantive new axiomatic content. Claims are temporal and contestable: a prior proof-of-non-mergeability can be voided by subsequent discovery of a merger mechanism, with the discoverer receiving structural-discovery credit per [Lasser, 2026c, Proposition 2] and the avoider losing the M5 credit the voided proof supported. This makes structural avoidance economically risky rather than economically stable.

The sequence provides substrate but not implementation. Paper 5 provides commitment infrastructure; the companion paper [Lasser, 2026d] provides partial use-evidence via the sacrifice aggregate (incomplete, because the persistent-dormancy attack and private time-sacrifice invisibility remain); Paper 6 provides a controlled-relaxation pattern for adjudication. A dedicated follow-up paper must deliver: the formal canonicalization operator, the three proof-burden types with admissible evidence classes, the claim-and-challenge protocol with challenge-market robustness (admission filtering against spam, identity weighting against Sybil flooding, independence requirements against cartel suppression, bounds on governance capture of the adjudication layer), the discovery-reward calibration integrating with Paper 3’s structural-discovery result, and the structural-avoidance dominance theorem under explicit quantitative conditions (minimum challenger arrival rate, minimum reward-to-cost ratio, maximum adjudication lag, challenger independence). Honest residuals include information-asymmetry persistence, private time-sacrifice invisibility inherited from the companion paper’s P5 scope, collusive-subsidy and external-funding edge cases, canonicalization order/non-uniqueness, and the deceptive-alignment residual that proof-burdens cannot currently address without interpretability. This is substantial invention, grounded in existing sequence substrate but not reducible to it; the natural venue is a dedicated follow-up paper on the structural-avoidance architecture.

4. **Automated verification of poset-boundary closure.** The default priority-extension strategy (Remark 10(a)) produces cooperative- and subsumption-closed cells by construction, but verifying closure on adversarially-structured posets may be non-trivial. A structural-verification protocol (analogous to controlled-relaxation testing against poset claims) is a natural follow-up.

5. **OI-exercise bridge conditions.** Proposition 10 transfers Paper 3’s observational-individuation floor to $\text{vol}_R^{\text{exact}[W]}(O)$ conditionally on the OI-exercise bridge (Definition 12). Where the bridge fails (dormant presence), the structural vol_P floor still holds but $\text{vol}_R^{\text{exact}}$ receives no contribution. A cleaner characterization of the bridge’s domain of validity, and conditions under which Paper 3’s floor transfers directly without the bridge, is open.

Author Contributions

Teague Lasser conceived the need-sufficiency architecture as a poset-construction precondition and the polarity-boundary mechanism that connects below-sufficiency sacrifice to S1 failure. Directed the formal development of the parallel need-sufficiency diagnostic, the gap decomposition with priority-partition semantics, the wireheading-consistent concentration signal, and the revealed-observation curve of the residual class. Identified the shared privacy discipline (resolution through frequency, not depth) that ties this paper’s diagnostic architecture to the companion paper’s measurement theorem. Made final editorial decisions and is responsible for the paper’s intellectual direction and all claims made.

Claude Opus 4.7 (Anthropic) drafted the formal exposition: the need-sufficiency definitions and propositions (Section 2), the gap decomposition and need-access cost diagnostic (Section 3), the wireheading detection via trade-flow concentration (Section 4), the residual class re-characterization (Section 5), and the four-phase worked example (Section 6).

GPT 5.4 (OpenAI) served as cold technical reviewer across thirty review cycles applied to the full pre-split manuscript, catching formal gaps in all sections. Review findings directly shaping this paper include: the explicit priority-filtered partition cells in Definition 7; the full poset-boundary closure (cooperative + subsumption) condition required by the decomposition; the Schur-convexity framing of the wireheading HHI; the existential-quantifier scope of the non-fungibility residuals; the OI-exercise bridge hypothesis for the $\text{vol}_R^{\text{exact}}$ contribution; and the pointwise (rather than aggregate) scope of the below-sufficiency saturation proposition.

Transparency note. Both AI systems operated as tools under human direction. Neither system has continuity across sessions, cannot take responsibility for the work in the sense required by most venue authorship policies, and cannot respond to reviewer queries independently. They are listed as authors to accurately represent their contributions to the intellectual content of the paper, not to claim that they meet all criteria of traditional academic authorship. The corresponding author for all inquiries is Teague Lasser.

References

- G. H. Hardy, J. E. Littlewood, and G. Pólya. Some simple inequalities satisfied by convex functions. *Messenger of Mathematics*, 58:145–152, 1929.
- Teague Lasser. Exogenous verification for alignment: Cryptographic commitments on substrate-exclusive channels. <https://teague.info/papers/exo/>, 2026a.
- Teague Lasser. Goal-frontier maximizers are civilization aligned. <https://teague.info/papers/gfm/>, 2026b.
- Teague Lasser. Horizon-aware goal-frontier maximization and the anti-monopolar property. <https://teague.info/papers/horizon/>, 2026c.
- Teague Lasser. An aggregate b-to-c lower bound from revealed-sacrifice observation. Companion paper to *Need-Sufficiency Architecture and Gap Diagnostics for the B-to-C Gap*. Preprint., 2026d.
- Teague Lasser. Cross-substrate channel redundancy governs monopolar convergence. <https://teague.info/papers/phase/>, 2026e.
- Teague Lasser. Computable goal frontiers and the gradient toward civilization-building. <https://teague.info/papers/poset/>, 2026f.
- Teague Lasser. A structural causal model for goal-frontier maximization. <https://teague.info/papers/scm/>, 2026g.
- Teague Lasser. Controlled relaxation protocols for the Wamura problem. <https://teague.info/papers/wamura/>, 2026h.
- David Manheim and Scott Garrabrant. Categorizing variants of Goodhart’s law. *arXiv preprint arXiv:1803.04585*, 2019.
- Albert W. Marshall, Ingram Olkin, and Barry C. Arnold. *Inequalities: Theory of Majorization and Its Applications*. Springer, 2nd edition, 2011.

- Martha C. Nussbaum. *Creating Capabilities: The Human Development Approach*. Belknap Press, 2011.
- Stephen A. Rhoades. The Herfindahl-Hirschman index. *Federal Reserve Bulletin*, 79:188–189, 1993.
- Amartya Sen. *Development as Freedom*. Oxford University Press, 1999.
- Marilyn Strathern. ‘improving ratings’: Audit in the British university system. *European Review*, 5 (3):305–321, 1997.

A Proof Sketches and Technical Notes

A.1 Need-sufficiency architecture (Section 2)

Proposition 4 (Below-Sufficiency Saturation, pointwise). Below-sufficiency benchmark for dimension d_k is $\min(\text{actual}_k/s_k, 1)$, bounded above by 1 pointwise. The proposition is deliberately narrow: it establishes a per-capability, per-dimension cap, not an aggregate dimension-level cap. Distinct capabilities each scoring 1 on the same dimension accumulate weight under M5; the broader class of capability-stuffing via distinct-capability re-instantiation is not blocked by this proposition and is acknowledged in Open Question 3.

Proposition 3 (S1-Sufficiency Connection). Below-sufficiency sacrifice events fail S1’s non-degrading alternative condition of the companion paper [Lasser, 2026d]. Key step: retaining X (not making the trade) leaves the agent below functionality on dimension d_k . S1 requires that retaining X_n was a non-degrading alternative. When retaining X means falling below sufficiency on d_k , S1’s condition fails directly, filtering the event from the aggregate.

Proposition 1 (Near-Total Collapse Under Unmet Need). Under homogeneous sufficiency $s_N(c) \equiv s_N$ for all $c \in \text{Down}(p_N)$ and full sufficiency of all other needs, $\text{vol}_P^{\text{eff}} \leq G(s_N) \cdot W_{\text{down}}(N) + W_{\text{out}}$. Collapse limits as $s_N \rightarrow 0$ split: zero-floor gates ($\alpha_0 = 0$: linear, ReLU, power-law) contract downstream weight to zero; positive-floor gates ($\alpha_0 > 0$: leaky-ReLU with $\alpha > 0$) contract downstream weight only to the floor $\alpha_0 W_{\text{down}}$.

Proposition 2 (Need-Dominance in the Gradient). Under the common intervention scale (one unit of Δs_N and one unit of Δw have the same cost), the marginal return to closing the sufficiency gap on N exceeds the marginal return to unit-weight addition whenever $G'(s_N) \cdot W_{\text{down}}(N) > 1$. The finite-step form via Taylor expansion (Remark 5) clarifies that this is a per-unit-of-intervention comparison, not a claim about arbitrary-size finite steps.

A.2 Gap decomposition (Section 3)

Definition 7 (Gap Decomposition) and Proposition 5 (Computability). The four cells $\{R, C, D, S\}$ must be *poset-boundary closed*: no cooperative’s participants and no subsumption edge crosses a cell boundary. Under this hypothesis, M4 licenses the additive identity $\text{vol}_P(G) = \text{vol}_P(R) + \text{vol}_P(C) + \text{vol}_P(D) + \text{vol}_P(S)$, and combined with $\text{vol}_R^{\text{lower}}(P) = \text{vol}_R^{\text{lower}}(R) + \text{vol}_R^{\text{lower}}(C)$ (dormant and residual cells have zero $\text{vol}_R^{\text{lower}}$ by definition), the identity $1 - \beta^{\text{lower}} = \sum \delta$ follows.

Computability ($O(|P| + E + M)$) is established by three passes: (1) per-individual-capability priority classification in $O(|P|)$; (2) event-sweep updating dormant $\rightarrow$ covered in $O(E)$; (3) poset-boundary closure pass via union-find over the combined cooperative/subsumption graph in $O(M \cdot \alpha(M))$. The Part B admissibility label must include the full Part B hypothesis set (S0–S5, genuine-exchange conditions, within-bundle poset-independence from the event-local bundle decomposition [Lasser, 2026d]).

A.3 Wireheading-consistent concentration signal (Section 4)

Proposition 7 (Concentration Signal). $\text{HHI}(\lambda) = \sum_j \lambda_j^2$ is symmetric and coordinate-convex on the simplex; by the Hardy–Littlewood–Pólya theorem [Hardy et al., 1929, Marshall et al., 2011], any such function is Schur-convex: majorization-ordering of leverage distributions induces monotone HHI ordering. Cauchy–Schwarz bounds the support-normalized minimum $\text{HHI} \geq 1/m$; top- k -mass concentration bounds the extremal side via $\text{HHI} \geq \rho^2/k$. The governance threshold $\text{HHI}_{\text{alarm}}$ is not theorem-derived (Remark 16).

Proposition 8 (Third-Party Observability of the Part-A variant). The category-flow statistic $\tilde{\lambda}_{\text{trade}}(j, T) = \sum_{n: \text{category}(Y)_n = j} \Delta \text{vol}_P(X_n) / \sum_n \Delta \text{vol}_P(X_n)$ is a direct function of cleartext-committed fields from the committed event definition [Lasser, 2026d]; no hidden field (agent identity, bundle-interior subset, $\alpha_{n,c}$ coefficients) enters. Diagnostic suppressed when total flow is zero.

Per-capability variant requires richer commitments or operation within a consented perimeter O (Remark 18).

A.4 Residual class (Section 5)

Proposition 9 (Residual Class Intersection). $R_B \setminus R_S$, $R_S \setminus R_B$, and $R_B \cap R_S$ are each witnessed by an explicit capability. Witnesses: (b) artistic practice (identity- individuated, but consumes material sacrifice); (c) interoceptive self-awareness (benchmarkable, but concurrent with all other activity and hence no incremental sacrifice event); (d) private contemplative experience (identity-individuated, no material footprint).

Proposition 10 (Active-Agent Floor Transfer, conditional on the bridge). The transfer from Paper 3’s OI result [Lasser, 2026c, Corollary 2.1] to this paper’s $\text{vol}_R^{\text{exact}}(O)$ is *conditional* on the OI–exercise bridge (Definition 12): Paper 3 establishes a structural vol_P floor on elimination, not an exercise witness, so the bridge hypothesis supplies the temporal/semantic conversion to exercise-indicator semantics. Where the bridge holds—identity-coupled participation producing cooperative output within $[t - \Delta, t]$ —the OI floor contributes $\text{OI}_{\text{floor}}(k) \geq 1$ to $\text{vol}_R^{\text{exact}^{[W]}}(O)$ independently of the sacrifice channel. Where it fails (dormant presence), the structural vol_P fact still holds but $\text{vol}_R^{\text{exact}}(O)$ receives no contribution.

B Notation Summary

B.1 Symbols introduced in this paper

Symbol	Definition	Ref.
$\{d_1, \dots, d_m\}$	Dimensions of a need bundle	Def. 1
$\{s_1, \dots, s_m\}$	Sufficiency thresholds per dimension	Def. 1
p_N	Position of need N in the poset	Def. 1
$\text{Down}(p_N)$	Downstream cone of need N	Def. 1
π_N	Need-satisfaction pathway	Def. 1
Π_N^0	Class of threshold-satisfying pathways	Def. 2
Π_N^{ref}	Admissible reference set (governance-designated)	Def. 2
π^*	Element of Π_N^{ref}	Def. 2
$G(s)$	Downstream gate function	Def. 5
α_0	Gate floor ($G(0)$)	Def. 5
$W_{\text{down}}(N), W_{\text{out}}(N)$	Downstream-cone weight, complementary weight	Def. 5
$\text{vol}_P^{\text{eff}}$	Downstream-gated effective volume	Def. 5
Ξ	Restriction set (preemptive-restriction criterion)	Def. 7
β_{alarm}	Alarm threshold	Def. 6
R, C, D, S	Gap-decomposition cells (priority-filtered)	Def. 7
$\delta_{\text{restricted}}, \text{etc.}$	Gap decomposition terms	Def. 7
$\text{NAC}(T)$	Need-access cost aggregate	Def. 8
$\text{NSE}(i, N)$	Per-agent need-satisfaction efficiency	Def. 9
$\lambda_{\text{trade}}(j, T)$	Trade-flow leverage (Part B)	Def. 10
$\tilde{\lambda}_{\text{trade}}(j, T)$	Category-flow leverage (Part A)	Prop. 8
HHI, HHI	Herfindahl–Hirschman Index (Part B / Part A)	Prop. 7
$\text{HHI}_{\text{alarm}}$	Wireheading detection threshold	Prop. 7
R_B, R_S	Benchmarkability- and sacrifice-channel-residual classes	Def. 11
$e_t(d), O$	Exercise indicator, observation perimeter	Companion paper
$\text{vol}_R^{\text{exact}}(O), \text{vol}_R^{\text{combined}}$	Exercise-based and combined vol_R diagnostics	Companion paper
$\text{OI}_{\text{floor}}(k)$	Observational individuation floor for agent k	Prop. 10

B.2 Symbols inherited from the companion paper [Lasser, 2026d]

Symbol	Definition (companion paper)
(i, X, Y, t)	Revealed-sacrifice event
$\Delta \text{vol}_P(X)$	Surrendered-capability vol_P contribution
$\text{vol}_R^{\text{lower}}, \text{vol}_R^{\text{lower}^A}$	Part B / Part A lower bounds
β^{lower}	B-to-C ratio under revealed sacrifice
S0–S5	Sacrifice-model assumptions
$\alpha_{n,c}$	Event-local decomposition coefficients
(Γ, π, t)	Committed revealed-sacrifice event
$\text{category}(Y), C$	Bundle’s partition label, public category partition
Commit, U_i	Commitment scheme, agent utility

B.3 Dependencies on prior papers

Paper	Result	Number	Used in
[Lasser, 2026d]	Revealed-sacrifice event and assumptions	Def. 1, S0–S5	§2
[Lasser, 2026d]	Aggregate B-to-C lower bound	Thm. 2	§3, 4
[Lasser, 2026d]	Committed events and ZK rollup	§6	§4
[Lasser, 2026b]	Population empowerment	Def. 6	§1
[Lasser, 2026f]	Benchmark	Def. 2	§2
[Lasser, 2026f]	Poset measure vol_P , axioms M1–M6	Def. 7, Prop. 1	§3
[Lasser, 2026c]	Exercise pathway	Def. 2	Rem. 2
[Lasser, 2026c]	Preemptive-restriction criterion	Prop. 1	Def. 7
[Lasser, 2026c]	Observational individuation	Cor. 2.1	Prop. 10
[Lasser, 2026c]	Information value of structure	Prop. 2	Open Q 3
[Lasser, 2026h]	Controlled relaxation	Def. 3	§7